Artificial Intelligence Driven Meta-Analysis

Introduction to Meta Analysis



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FACULTY OF APPLIED SCIENCES

PHD OF ARTIFICIAL INTELLIGENCE EMPOWERED SMART HEALTHY AGEING

LEARNING MODULE OUTLINE

Academic Year	2026/2027	Semester	1
Module Code	AIHA8123		
Learning Module	Artificial Intelligence Driven Meta-Analysis		
Pre-requisite(s)	Nil		
Medium of Instruction	English		
Credits	3	Contact Hours	45 hrs
Instructor	Dr. Wenwang Rao	Email	raoww@mpu.edu.mo
Office	ACADEMIC BUILDING; E1303-2	Office Phone	8599 3826

Times	Assessment Activities	Weighting (%)
2	A1. Assignment / Classwork	25
1	A2. Tests	25
1	A3. Project	50



In-class assignments



The course module outline, PowerPoint slides, assignments, and other materials will be uploaded to Canvas. If you have any questions, please email raoww@mpu.edu.mo.

My office hours are at 10.00 am to 1.00 pm on Tuesday and on Thursday.

My office is currently located at the Academic Building E1303-2 (教研樓) or Room 206, Building 11, Dezhi Plaza, Macao-Hengqin International Education (University) Town.

Attendance

- Students are required to maintain a minimum attendance rate of 70%.



- For Chinese students, full Chinese/English names are required;
- For Foreign students, full names or nicknames are acceptable;
- Attendance are based on the sign-in sheet only.







Reminder for Group Project

1. **Date of Final Report：** Dec. 5, 2025

- **Deadline of Submission：** 10.00 am

2. **Date of Presentation：** Dec. 12, 2025

- **Time：** 10:00-13:00 and 19:00-22:00
- **Deadline of Submission：** 10.00 am



E-mail: Dr. Rao: raoww@mpu.edu.mo

Major Requirements for Group Project

- Teams consist of **3 or 4 randomly** assigned members
- The project data will be released on **Sep 12, 2026** (today)
- Your team need to select **a topic** and utilize it to develop a detailed project.
- Teams are required to submit ***the final project reports with a Turnitin report, the ppt or pdf of the presentation slides, the work contribution and Gen AI documents***.
- Teams should select their **own presenter**.
- The project leader will submit **all project files**, and every member must submit a **peer evaluation for the entire team.**
- **For more details, please refer to the files in Canvas.**



Literature review:

- The general term for all attempts to synthesize the results and conclusions of two or more publications on a given topic .
- A review may or may not be systematic.



- **Narrative review**

Traditional expert review; usually subjective in nature.

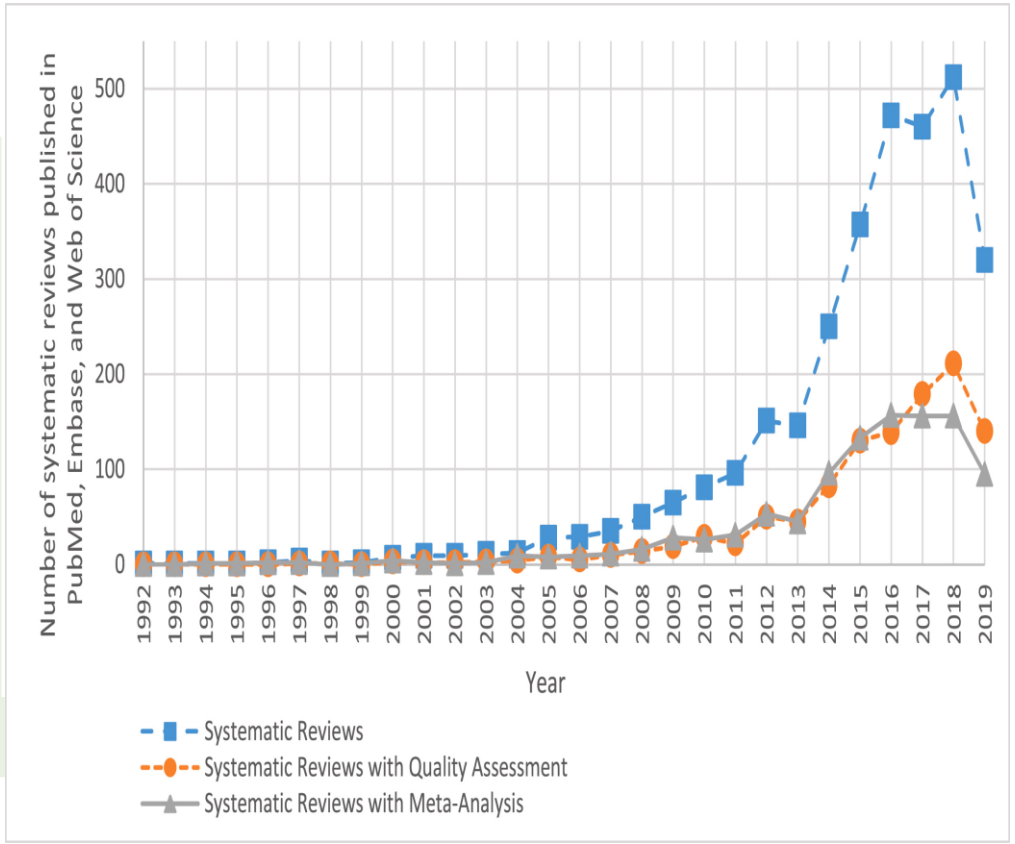
What is a Systematic Review?

- "A review that is conducted according to clearly stated, scientific research methods, and is designed to minimize biases and errors inherent to traditional, narrative reviews."



Systematic Review History:

- **Astronomers** claim to be the first users of this method.
- **Explosion** of SRs in health sciences in mid **1980s**.
- Term **"systematic review"** was coined by **health care researchers.**
- SR became a **significant tool** for" **evidence- based medicine" or " evidence- based practice"**
- Growth of SR s in health sciences.



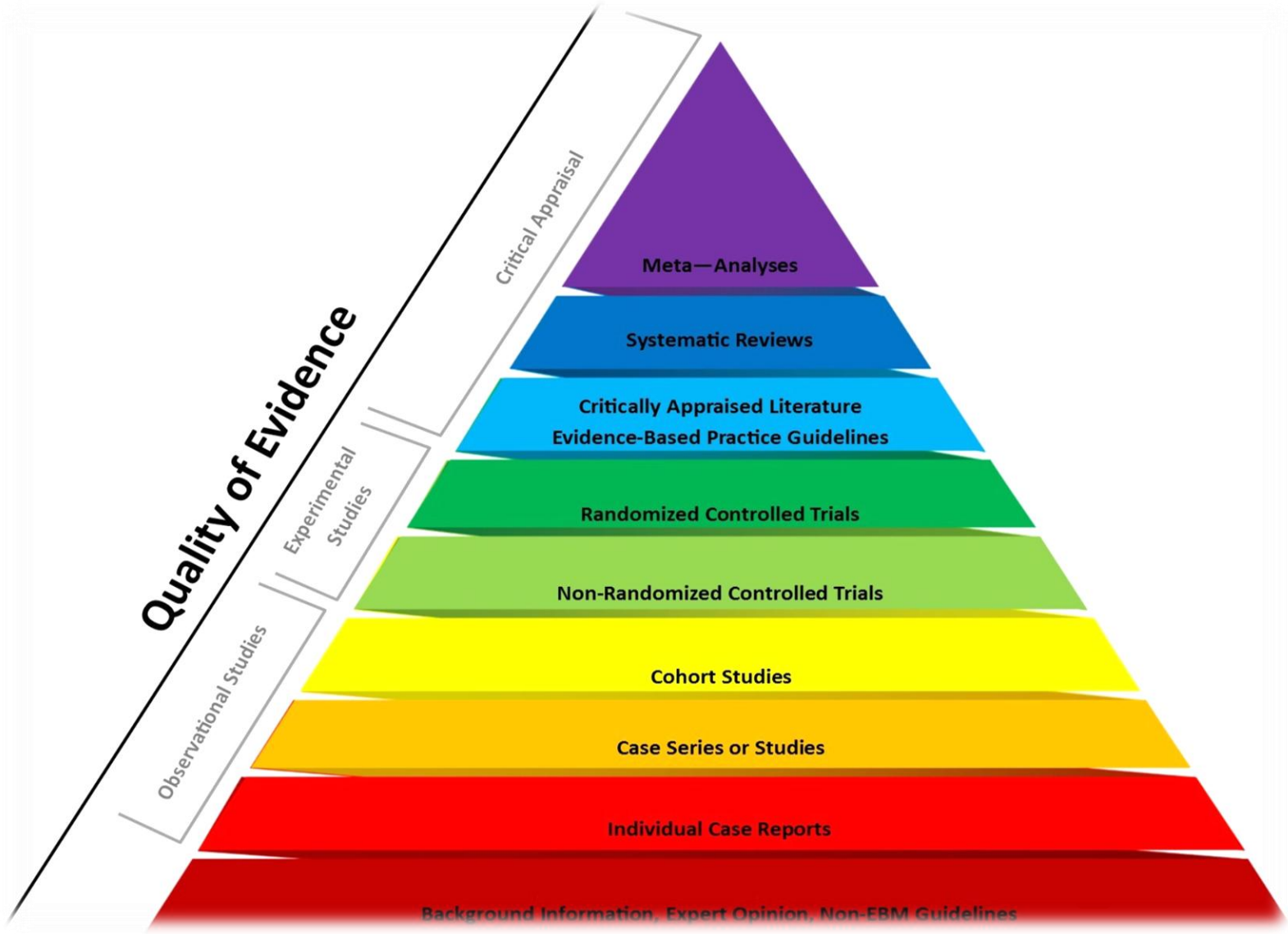
	Systematic review	Traditional narrative review
Deciding on review question	Start with clear question to be answered or hypothesis to be tested	May also start with clear question to be answered, but they more often involve general discussion of subject with no stated hypothesis
Searching for relevant studies	Strive to locate all relevant published and unpublished studies to limit impact of publication and other biases	Do not usually attempt to locate all relevant literature

	Systematic review	Traditional narrative review
Deciding which studies to include and exclude	Involve explicit description of what types of studies are to be included to limit selection bias on behalf of reviewer	Usually do not describe why certain studies are included and other excluded
Assessing study quality	Examine in systematic manner methods used in primary studies, and investigate potential biases in these studies and sources of heterogeneity	Often do not consider differences in study methods or study quality
Synthesizing study results	Base their conclusion on those studied which are most methodologically sound	Often do not differentiate between methodologically sound and unsound

Systematic review vs (Traditional) Literature review

	Systematic Review	Literature Review
Definition	High-level overview of primary research on a focused question that identifies, selects, synthesizes, and appraises all high quality research evidence relevant to that question	Qualitatively summarizes evidence on a topic using informal or subjective methods to collect and interpret studies
Goals	Answers a focused clinical question Eliminate bias	Provide summary or overview of topic
Question	Clearly defined and answerable clinical question Recommend using PICO as a guide	Can be a general topic or a specific question
Components	Pre-specified eligibility criteria Systematic search strategy Assessment of the validity of findings Interpretation and presentation of results Reference list	Introduction Methods Discussion Conclusion Reference list
Number of Authors	Three or more	One or more
Timeline	Months to years Average eighteen months	Weeks to months
Requirement	Thorough knowledge of topic Perform searches of all relevant databases Statistical analysis resources (for meta-analysis)	Understanding of topic Perform searches of one or more databases

Hierarchy of scientific evidence



What is the significance of Systematic Reviews?

- The large amount of medical literature requires clinicians and researchers alike to rely on systematic reviews in order to make an informed decision.
- Systematic Reviews minimize bias.
- "A systematic review is a more scientific method of summarizing literature because specific protocols are used to determine which studies will be included in the review."

Why are Systematic Reviews Necessary?

- "The volume of published material makes it impractical for an individual clinician to remain up to date on a variety of common conditions.
- This is further complicated when individual studies report conflicting conclusions, a problem that is prevalent when small patient samples and retrospective designs are used."



Key Characteristics of Systematic Reviews

- Clearly stated title and set of objectives with predefined eligibility criteria for studies.
- Comprehensive strategy to search for relevant studies (unpublished and published)
- Assessment of the validity of the findings of the included studies.



Key Characteristics of Systematic Reviews

- Explicit and justified criteria for the inclusion or exclusion of any study.
- Clear presentation of characteristics of each study included and an analysis of methodological quality
- Comprehensive list of all studies excluded and justification for exclusion.

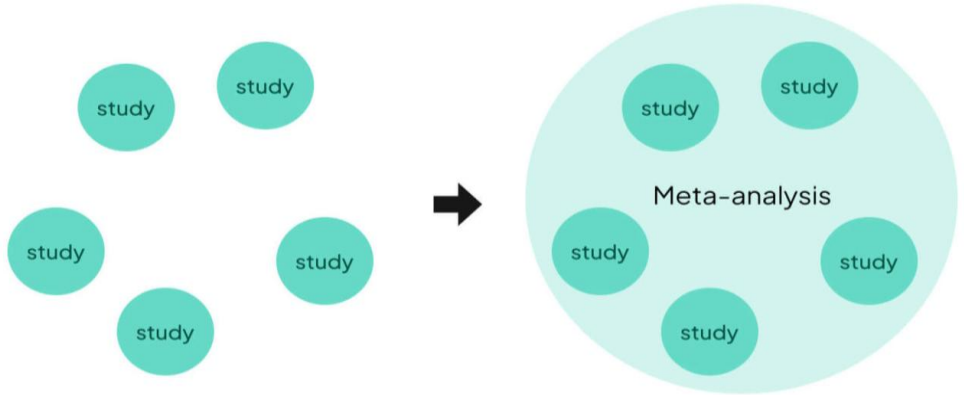
Key Characteristics of Systematic Reviews

- Clear **analysis** of the **results** of the eligible studies
  - –**statistical synthesis** of data **(meta-analysis)** if appropriate and possible;
  - –or **qualitative synthesis**
- **Structured report** of the review **clearly stating** the **aims**, describing the **methods** and **materials** and reporting the **results.**



What is Meta-Analysis?

- Analysis of Analyses
- An Objective and Quantitative Methodology for synthesizing previous studies and research on a particular topic into an overall finding.
- Meta-Analysis allows us to **collect, code, compare or combine** results from different studies and **interpret** using **statistical methods** facilitating statistically guided **decisions about the strength of observed effects** and the **reliability** of results across a range of studies.



What is a Meta Analysis?

Meta-analysis

- DEFINITION
- EXAMPLES
- BENEFITS
- WEAKNESSES

Meta-analysis is a research method that combines the results of many different studies on the same topic to reach a stronger and more reliable conclusion.



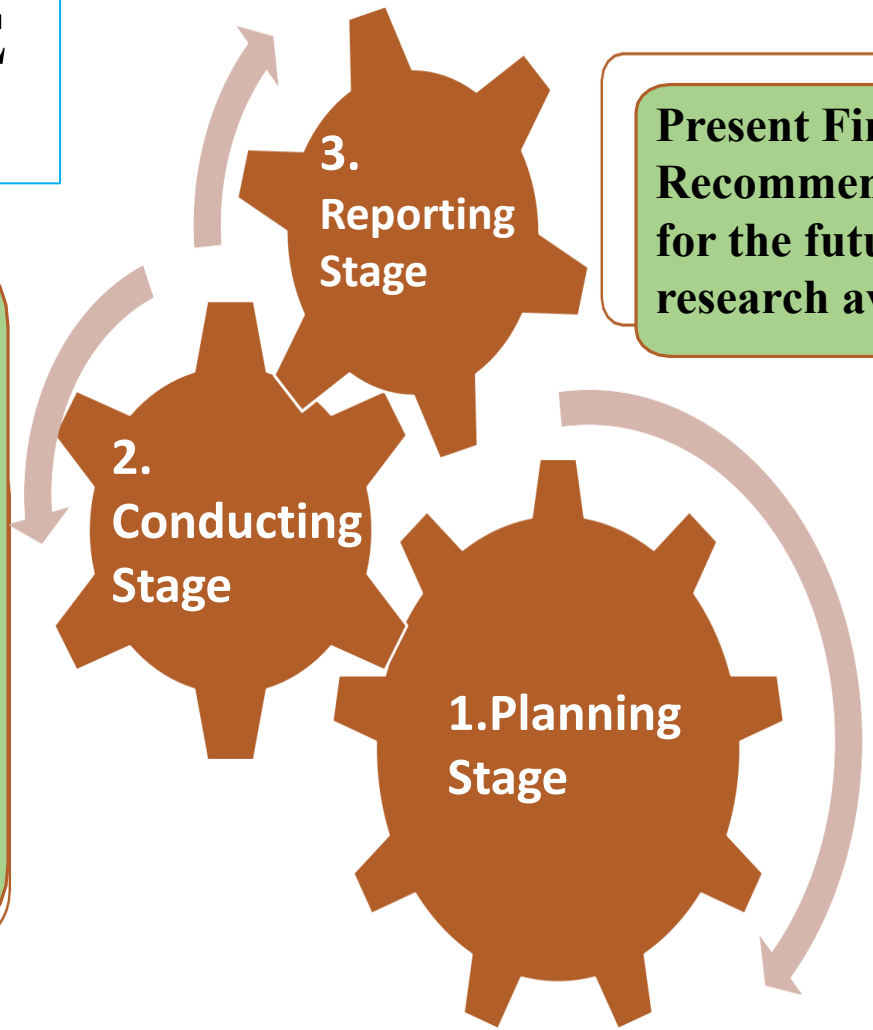
Why Meta-Analysis?

- To provide a **Summary effect** of all previous studies
- **Overall conclusion** of various studies
- Efficient method for **claiming Gaps in Model/Theory**
- Tool for the **Theory Development**



SYSTEMATIC LITERATURE REVIEW

1. Search Term and Criteria Selection
2.Exclusion Criteria
3.Assessment of Review Team
4.Final List of Studies
5.Analysis and
6.Synthesis of Literature

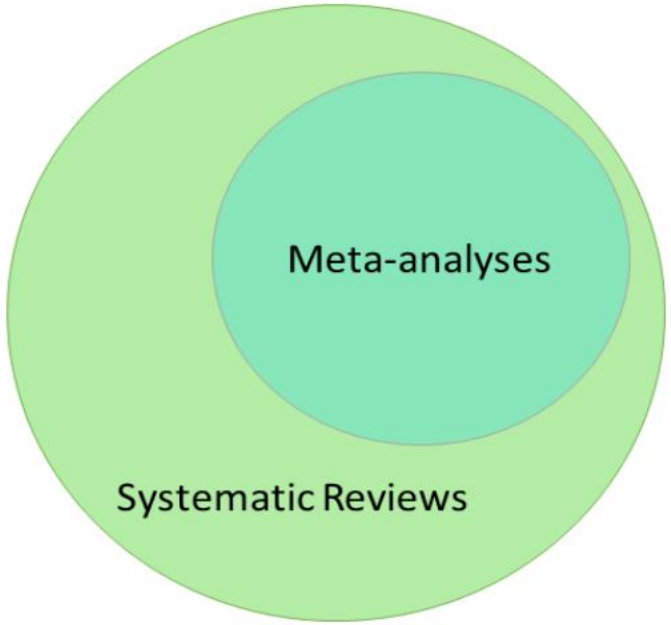


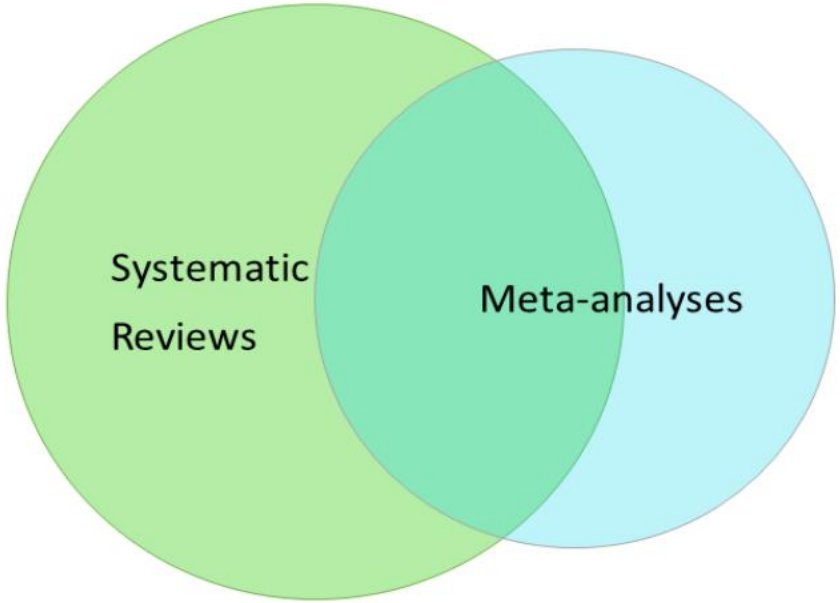
Present Findings Recommendations for the future research avenues

1.Selection of Review Team
2.Exploring the relevant Literature

Relationship between SR and MA

- SR is the process of undertaking the reviews, and MA may be one outcome.
- Not all systematic reviews include meta-analysis, but all meta-analysis are found in systematic reviews.



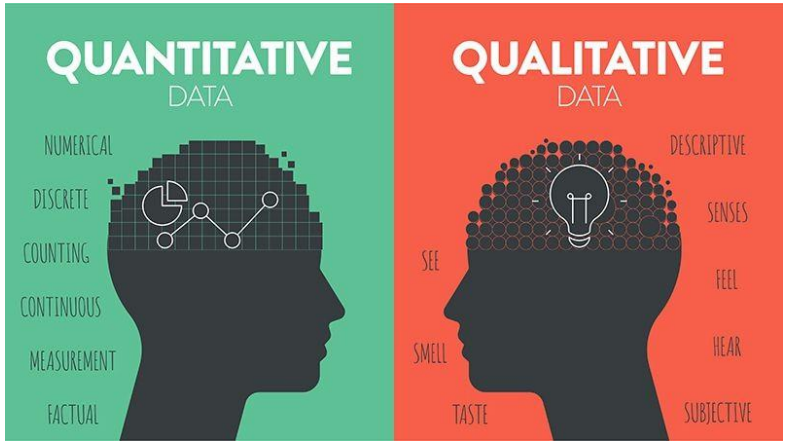


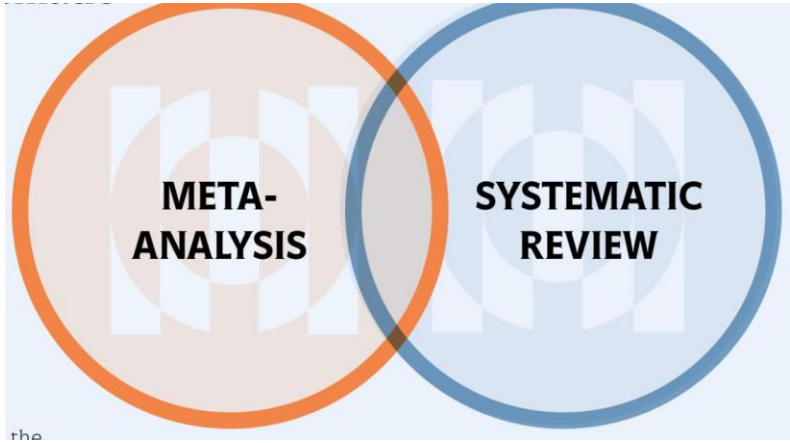
What is PRISMA?

PRISMA

- A set of **guidelines** to **improve reporting** systematic reviews and meta-analyses
- Ensures **transparency** and **completeness**







PRISMA: Preferred Reporting Items for Systematic reviews and Meta-Analyses



Prof. Joanne McKenzie



Dr. Matthew Page

Welcome to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) website

Here you can access information about the PRISMA reporting guidelines, which are designed to help authors transparently report why their systematic review was done, what methods they used, and what they found.

The main PRISMA reporting guideline (the PRISMA 2020 statement) primarily provides guidance for the reporting of systematic reviews evaluating the effects of interventions. PRISMA 2020 is complemented by various PRISMA extensions, which provide guidance for the reporting of different types or aspects of systematic reviews and other types of evidence synthesis (e.g. scoping reviews).

Development, updating and implementation of the PRISMA reporting guidelines is overseen by the PRISMA Executive, which is currently co-chaired by Prof Joanne McKenzie and Dr Matthew Page at Monash University.

Key documents

PRISMA 2020 checklist

PRISMA 2020 flow diagram

PRISMA 2020 statement paper

PRISMA 2020 Explanation and Elaboration paper

PRISMA

https://www.prisma-statement.org/

What is PRISMA guidelines?

• The PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines are a set of standardized reporting criteria aimed at improving the quality and transparency of systematic reviews and meta-analyses. Here's a breakdown of what they include:

• PRISMA 2020 Statement: This is the latest update of the guidelines. It consists of a 27-item checklist that authors should follow to ensure that all essential aspects of a systematic review are reported. The items cover everything from the title of the review to the funding sources.

• PRISMA Flow Diagram: This is a visual representation of the flow of information through the different phases of a systematic review. It shows the number of records identified, included, and excluded, along with reasons for exclusions.

What is PRISMA guidelines?

• PRISMA Extensions: These are additional guidelines tailored for specific types of reviews or aspects of reporting. Some examples include PRISMA-P (for protocols), PRISMA-ScR (for scoping reviews), and PRISMA-Children (for systematic reviews that include children's data

• Explanation and Elaboration Document: This document provides detailed guidance and examples for each checklist item, helping authors understand how to properly implement the guidelines. Availability: PRISMA guidelines are freely available online and are widely used by researchers to improve the rigor and reproducibility of their systematic reviews.

Website: https://www.prisma-statement.org/

PRISMA

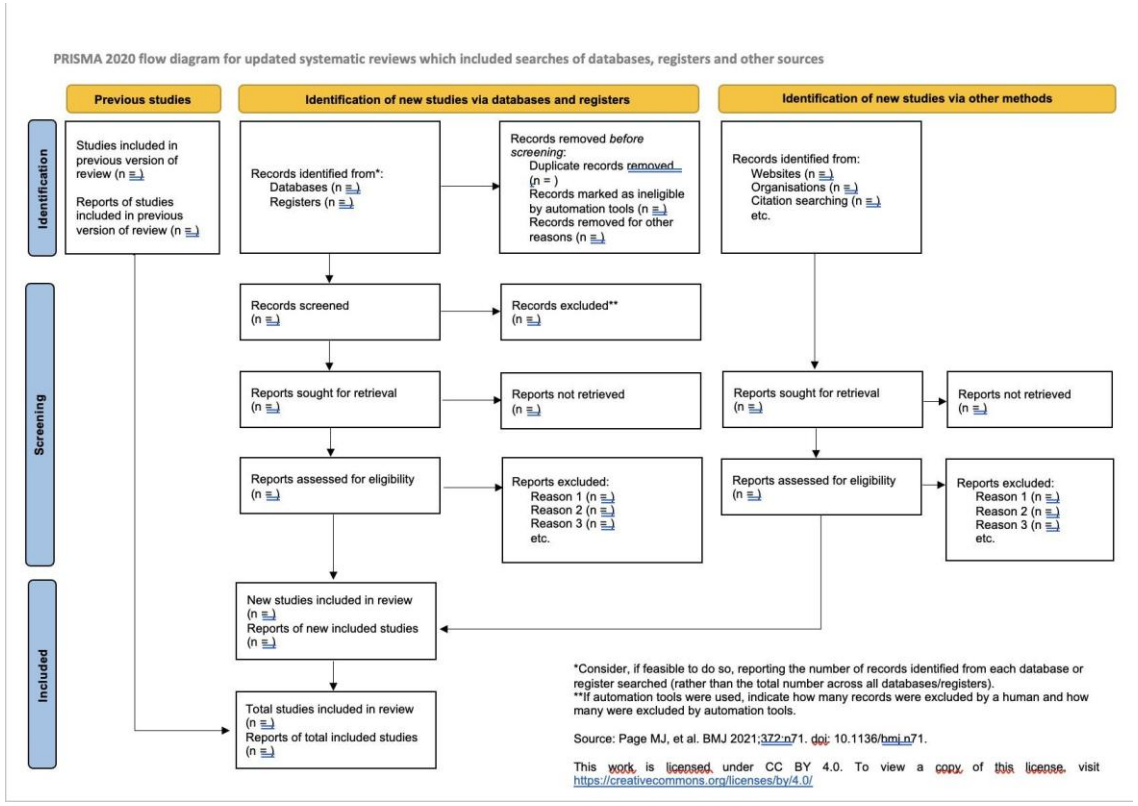
PRISMA components

Checklist

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	

Flow diagram



Statement

The PRISMA 2020 statement: an updated guideline for reporting systematic reviews

BMJ 2021 ; 372 doi: https://doi.org/10.1136/bmj.n71 (Published 29 March 2021)
Cite this as: *BMJ* 2021;372:n71

But how to conduct a systematic review and meta analysis step by step?

1. Research question:

- Should be feasible, interesting, novel, ethical, and relevant.
- Therefore, a clear, logical, and well-defined research question should be formulated.
- Usually, two common tools are used: PICO or SPIDER.

- **PICO** (Population, Intervention, Comparison, Outcome) is used mostly in quantitative evidence synthesis.

- **SPIDER** (Sample, Phenomenon of Interest, Design, Evaluation, Research type) was proposed as a method for qualitative and mixed methods search.

2. Preliminary search: Validate idea, see if it has been done previously. Propose no. of included studies

3. Inclusion & Exclusion criteria:

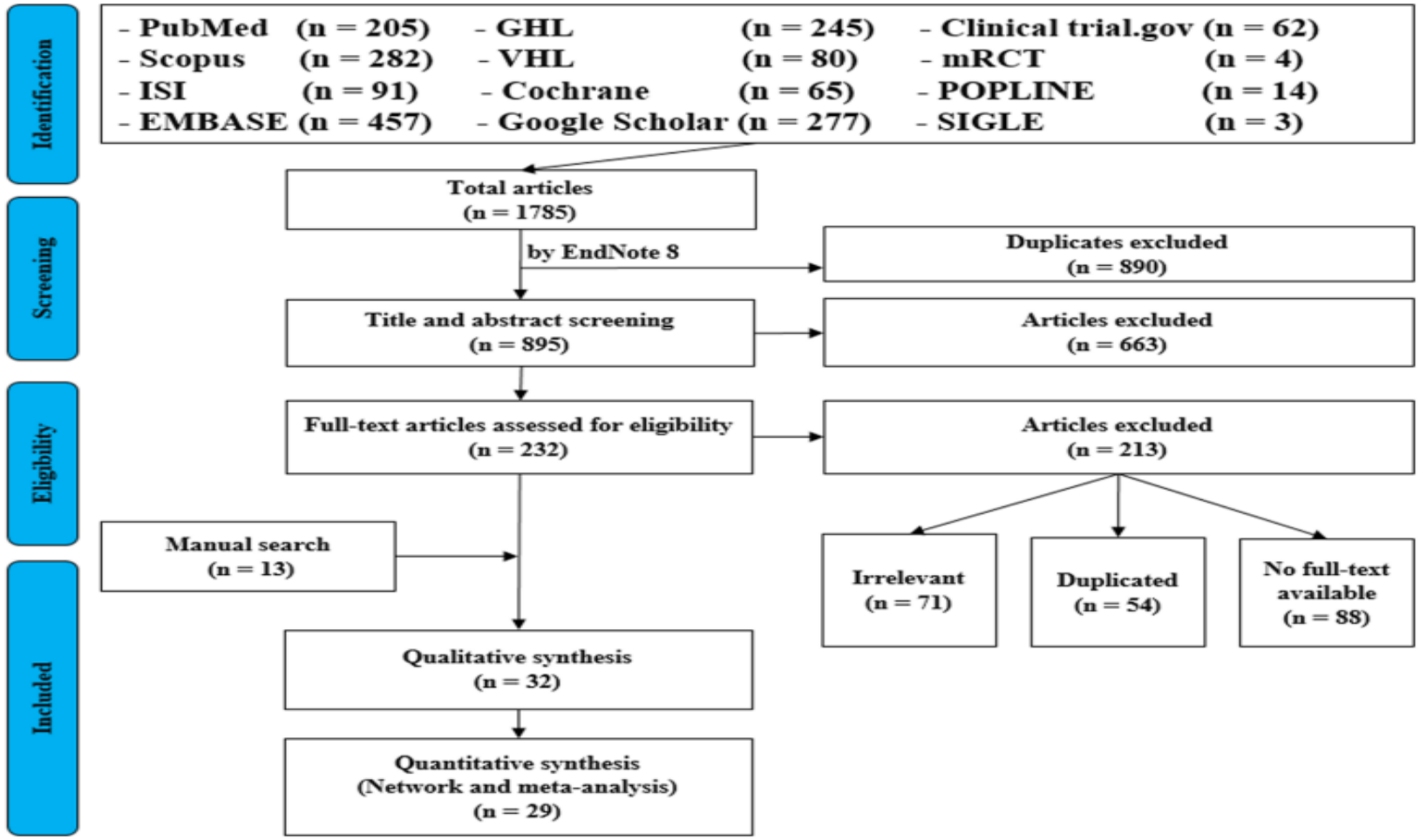


Fig. 2 PRISMA flow diagram of studies' screening and selection

4. Search strategy & 5. Searching databases:

- PubMed, EMBASE, Google Scholar, Scopus, Cochrane etc. According to AMSTAR guidelines, at least two databases have to be searched.
- Boolean operators, such as “AND”, “OR”, “NOT” are used for refining search strategy.

6. Protocol writing & registration:

- Protocol registration at an early stage **guarantees transparency** in the research process and **protects from duplication problems.**
- Besides, it is considered a documented proof of team plan of action, research question, eligibility criteria, intervention/exposure, quality assessment, and pre-analysis plan.
- Researchers should send it to the principal investigator (PI) to revise it, then upload it to **registry sites** [Proposed by Cochrane and Campbell collaborations; PROSPERO etc.]

7. Title and abstract screening:

- Reviewers (2-3) decide to include or exclude any report based on criteria.

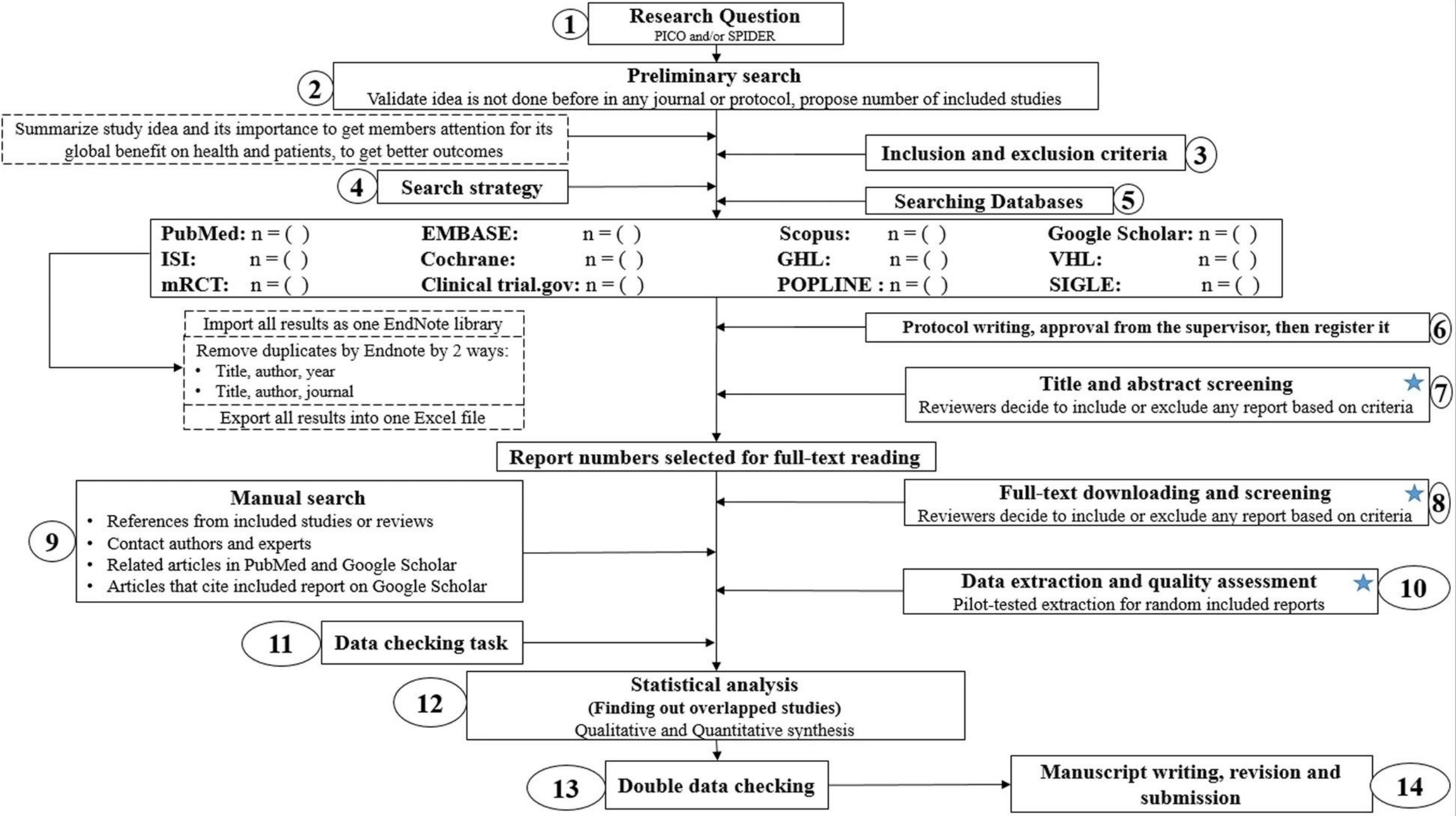
8. Full text downloading and screening

9. Manual search:

- Searching references from included studies/reviews
- Contacting authors and experts, and
- Looking at related articles/cited articles in PubMed and Google Scholar.

10. Data extraction and Quality assessment: (More on quality assessment and risk of bias assessment later on)

11. Statistical analysis (Meta analysis) → **12. Manuscript writing, revision & Submission**


1
Research Question
PICO and/or SPIDER
2
Preliminary search
Validate idea is not done before in any journal or protocol, propose number of included studies
Summarize study idea and its importance to get members attention for its global benefit on health and patients, to get better outcomes
Inclusion and exclusion criteria
3
4
Search strategy
Searching Databases
5
PubMed: n = ( )
EMBASE: n = ( )
Scopus: n = ( )
Google Scholar: n = ( )
ISI: n = ( )
Cochrane: n = ( )
GHL: n = ( )
VHL: n = ( )
mRCT: n = ( )
Clinical trial.gov: n = ( )
POPLINE : n = ( )
SIGLE: n = ( )
Import all results as one EndNote library
Remove duplicates by Endnote by 2 ways:
• Title, author, year
• Title, author, journal
Export all results into one Excel file
Protocol writing, approval from the supervisor, then register it
6
Title and abstract screening
Reviewers decide to include or exclude any report based on criteria
7
Report numbers selected for full-text reading
Full-text downloading and screening
Reviewers decide to include or exclude any report based on criteria
8
Manual search
• References from included studies or reviews
• Contact authors and experts
• Related articles in PubMed and Google Scholar
• Articles that cite included report on Google Scholar
9
Data extraction and quality assessment
Pilot-tested extraction for random included reports
10
11
Data checking task
12
Statistical analysis
(Finding out overlapped studies)
Qualitative and Quantitative synthesis
13
Double data checking
Manuscript writing, revision and submission
14

Quality Assessment and Risk of Bias

- However, well planned the systematic review or meta-analysis is, if the quality of evidence in the studies is low, the quality of the meta-analysis decreases and incorrect results can be obtained.

- The quality of the studies included in the systematic review determines the certainty with which conclusions can be drawn.

- **Quality assessment** is the assessment of the inclusion of methodological safeguards within a study whereas **Risk of bias assessment** concerns the implication of the inclusion of such safeguards for study results.

- Many a times these two terms (quality assessment and risk of bias assessment) is used interchangeably.

Quality Assessment and Risk of Bias

- Once all the relevant studies have been identified, the studies should undergo a quality assessment. This is particularly important if there is contradictory evidence.

- Even when using randomized studies with a high quality of evidence, evaluating the quality of evidence precisely helps determine the strength of recommendations in the meta-analysis.

- Various tools have been designed to check quality assessment



- The **Jadad score (Oxford Quality Rating scale)** is frequently used for quality assessment of RCTs

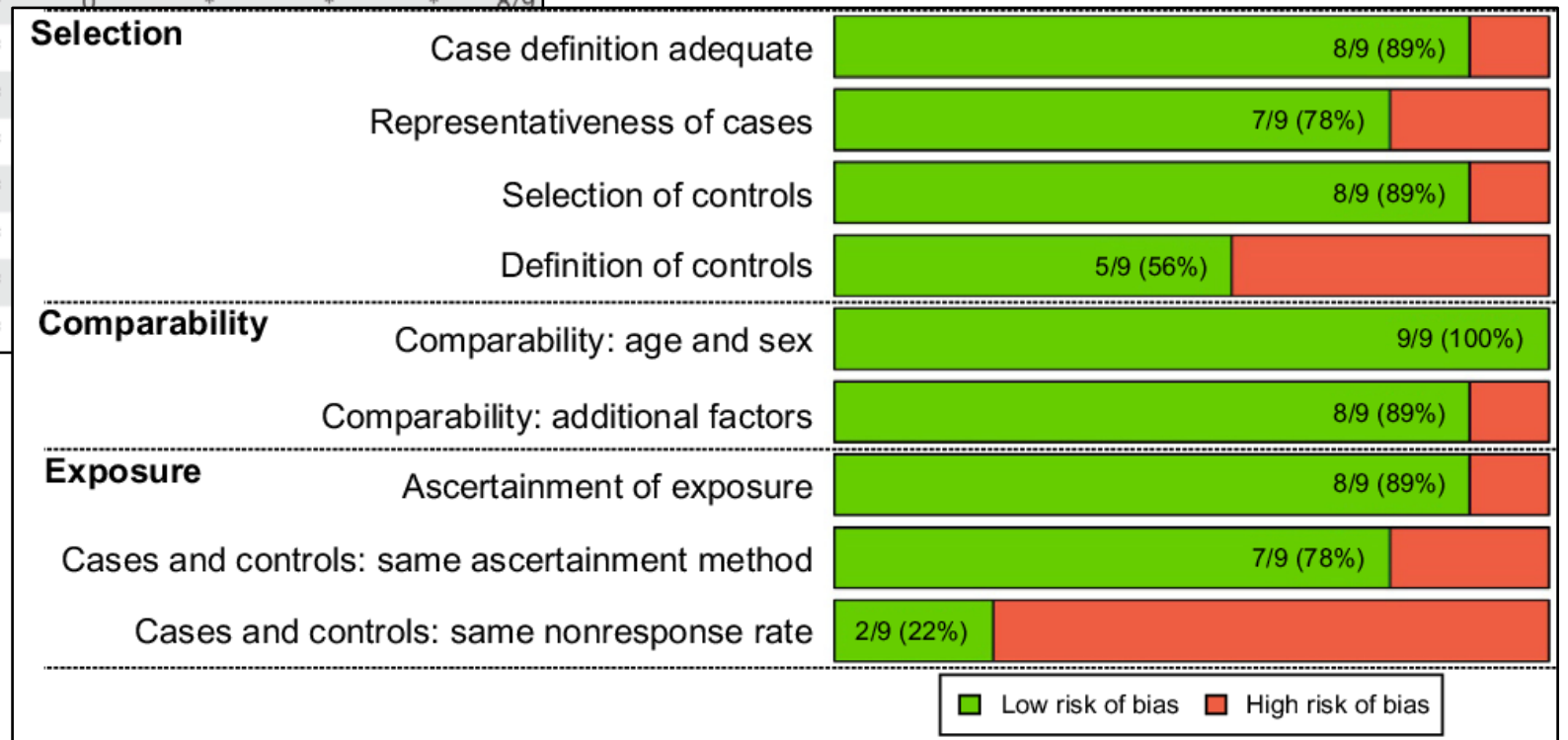
Item	Maximum Points	Description
Randomization	2	1 point if randomization is mentioned 1 additional point if the method of randomization is appropriate Deduct 1 point if the method of randomization is inappropriate
Blinding	2	1 point if blinding is mentioned 1 additional point if the method of blinding is appropriate Deduct 1 point if the method of blinding is inappropriate
Withdrawals	1	1 point if the number and the reasons for withdrawal in each group are stated

Eight items	Answer	Score
Was the study described as randomized?	Yes	+1
	No	0
Was the method of randomization appropriate?	Yes	+1
	No	-1
	Not described	0
Was the study described as blinding? [a]	Yes	+1
	No	0
Was the method of blinding appropriate?	Yes	+1
	No	-1
	Not described	0
Was there a description of withdrawals and dropouts?	Yes	+1
	No	0
Was there a clear description of the inclusion/exclusion criteria?	Yes	+1
	No	0
Was the method used to assess adverse effects described?	Yes	+1
	No	0
Was the methods of statistical analysis described?	Yes	+1
	No	0

a: double-blind got 1 score, single-blind got 0.5 score.

- **The Newcastle-Ottawa score** is used for nonrandomized studies

Study	Selection				Comparability		Outcome			Total (9/9)
	Repre-sentative of the exposed cohort	Selec-tion of external control	Ascertain-ment of exposure	Outcome of interest not present at the start of the study	Comparability of cohorts: Main factor	Comparability of cohorts: Additional factor	Assessment of outcomes	Sufficient follow-up time[a]	Adequacy of follow-up	
Van Steenberghe et al[50]	*	0	*	*	*	0	*	*	*	7/9
Sanna et al[51]	0	*	*	*	*	0	*	*	*	7/9
Johansson et al[52]	*	0	*	*	*	0	*	*	0	6/9
Meloni et al[53]	*	0	*	*	*	0	*	*	*	7/9
Nikzad et al[54]	*	0	*	*	*	*	*	*	*	7/9
Van de Velde et al[55]	*	*	*	*	*	0	*	*	*	8/9
Horwitz et al[56]	0	0	*	*	*					
Komiyama et al[57]	0	0	*	*	*					
Pozzi et al[58]	*	0	*	*	*					
D'Haese et al[59]	0	*	*	*	*					
Landazuri-Del Barrio et al[60]	*	0	*	*	*					
Marra et al[61]	0	0	*	*	*					
Browaeys et al[62]	0	*	*	*	*					



Risk of Bias

- The study limitations are evaluated using the “risk of bias” method proposed by Cochrane.

- The risk of bias is defined as the risk of systematic error or a deviation from reporting the truth or an appropriate evidence finding.

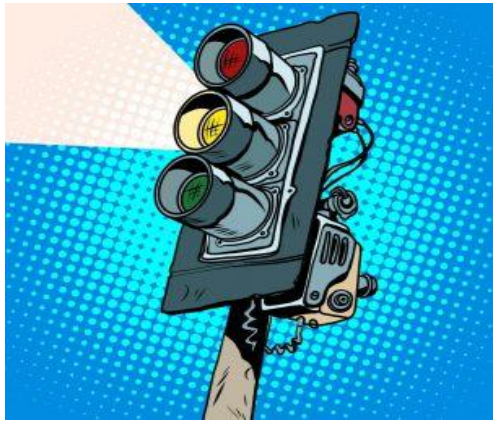
- This method classifies bias in **randomized studies** as “low,” “high,” or “unclear” on the basis of the presence or absence of six processes (Random sequence generation, Allocation concealment, Blinding participants or investigators, Incomplete outcome data, Selective reporting, and Other biases)

- Again, there are number of tools to assess risk of bias (according to different kind of studies)

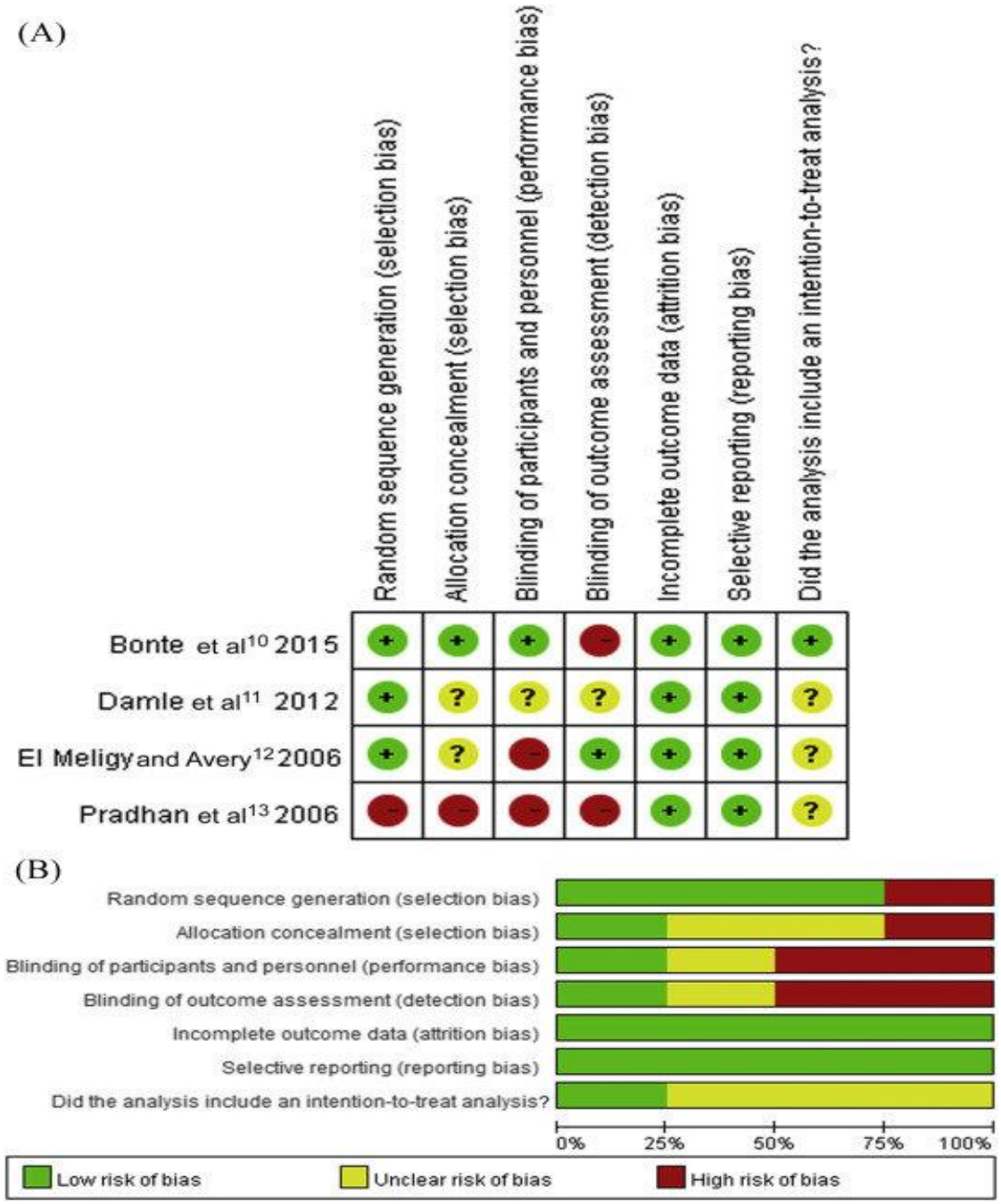
Table 1. The Cochrane Collaboration's Tool for Assessing the Risk of Bias [28]

Domain	Support of judgement	Review author's judgement
Sequence generation	Describe the method used to generate the allocation sequence in sufficient detail to allow for an assessment of whether it should produce comparable groups.	Selection bias (biased allocation to interventions) due to inadequate generation of a randomized sequence.
Allocation concealment	Describe the method used to conceal the allocation sequence in sufficient detail to determine whether intervention allocations could have been foreseen in advance of, or during, enrollment.	Selection bias (biased allocation to interventions) due to inadequate concealment of allocations prior to assignment.
Blinding	Describe all measures used, if any, to blind study participants and personnel from knowledge of which intervention a participant received.	Performance bias due to knowledge of the allocated interventions by participants and personnel during the study.
	Describe all measures used, if any, to blind study outcome assessors from knowledge of which intervention a participant received.	Detection bias due to knowledge of the allocated interventions by outcome assessors.
Incomplete outcome data	Describe the completeness of outcome data for each main outcome, including attrition and exclusions from the analysis. State whether attrition and exclusions were reported, the numbers in each intervention group, reasons for attrition/exclusions where reported, and any re-inclusions in analyses performed by the review authors.	Attrition bias due to amount, nature, or handling of incomplete outcome data.
Selective reporting	State how the possibility of selective outcome reporting was examined by the review authors, and what was found.	Reporting bias due to selective outcome reporting.
Other bias	State any important concerns about bias not addressed in the other domains in the tool. If particular questions/entries were prespecified in the review's protocol, responses should be provided for each question/entry.	Bias due to problems not covered elsewhere in the table.

Traffic light graph



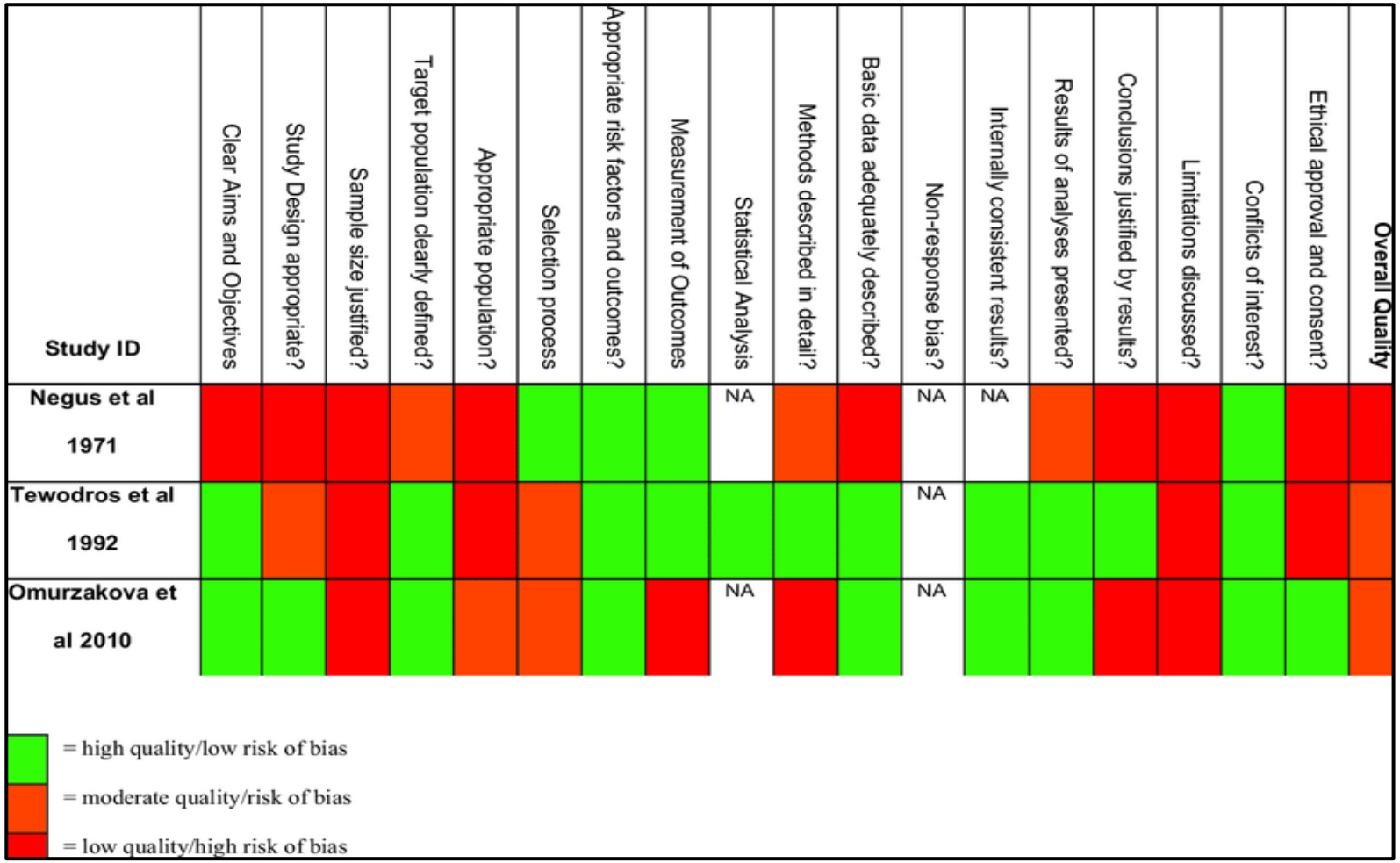
- Low risk of bias (**Green**)
- Unclear risk (**Orange/Yellow**)
- High risk of bias (**Red**)



Few other Risk of Bias assessment tools

- **AMSTAR 2:** A MeaSurement Tool to Assess systematic Reviews
- **GRADE:** Grading of Recommendations Assessment, Development and Evaluation
- **AXIS:** Appraisal tool for Cross-Sectional Studies

- **ROBIS:** Risk Of Bias in Systematic Reviews
- **NIH** checklist



Meta Analysis

- The **statistical methods for combining the results of a number of studies** are known as meta-analysis.

- The aim of a meta-analysis is to derive a conclusion with **increased power and accuracy** than what could not be able to achieve in individual studies.

- It should be emphasized that **not all systematic reviews will contain a meta-analysis**; this will depend on whether the systematic review has located studies that are sufficiently similar to make it reasonable to consider combining their results

- Therefore, before analysis, it is crucial to evaluate the direction of **effect***, size of effect, homogeneity of effects among studies, and strength of evidence.

- Thereafter, the data are reviewed qualitatively and quantitatively.

- If it is determined that the different research outcomes cannot be combined, all the results and characteristics of the individual studies are displayed in a table or in a descriptive form; this is referred to as a **qualitative review**.

- A **meta-analysis is a quantitative review**, in which the clinical effectiveness is evaluated by calculating the weighted pooled estimate for the interventions in at least two separate studies.

- The pooled estimate is the outcome of the meta-analysis, and is typically explained using a **Forest plot**

Effect Size

- SR/MA was primarily designed for RCT (Clinical trials)

- The meta-analysis result may show either a benefit or lack of benefit of a treatment approach that will be indicated by the effect size, which is the term used to describe the **treatment effect of an intervention.** Treatment effect is the gain (or loss) seen in the experimental group relative to the control group.

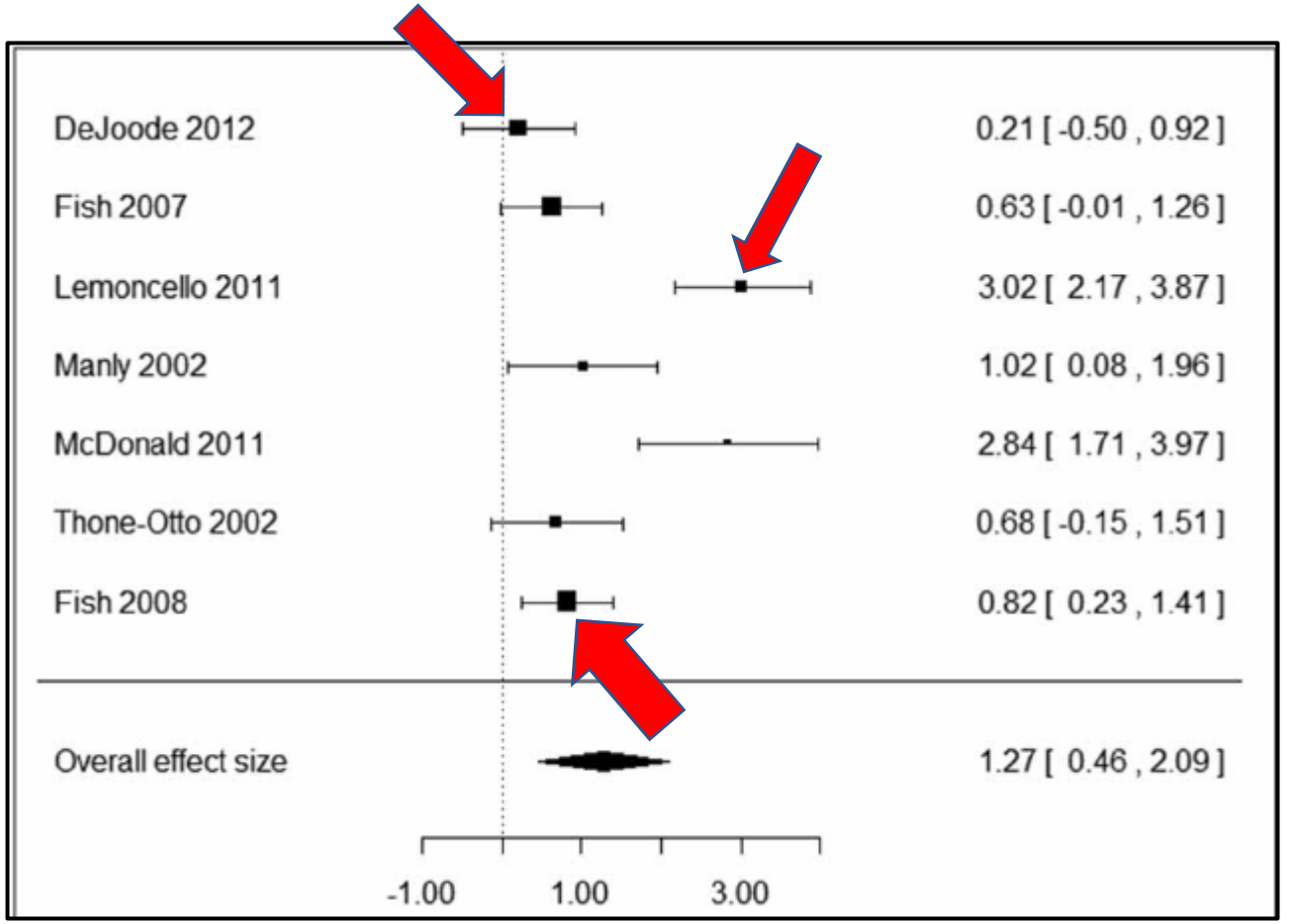
- Statistically speaking, Effect size is a measure of strength of relationship between two variables.

- Binary outcomes: Odds Ratio (OR), Relative Risk (RR)
- Continuous outcomes: Mean Difference (MD), Standardized Mean Difference (SMD)

- In other words, Effect size is a **dimensionless estimate** (i.e., a measure with no units) that indicates both **direction and magnitude of the treatment effect**

Magnitude and direction depends upon:

- Sample size
- Variance
- Reliability of outcome measures



Heterogeneity

- Heterogeneity simply means **variability among studies.**
- Heterogeneity tells us, are these studies different? If yes, can we quantify it? Should these studies be combined? If yes, how?

- Different types of heterogeneity:

A. **Clinical Heterogeneity**: Difference in study methods that affect the ability to compare and/or combine data from different studies. E.g., participant demographics, risk or severity of disease, study settings, frequency and intensity of intervention and how outcomes were measured

B. **Methodological Heterogeneity:** Risk of bias assessment

C. **Statistical Heterogeneity:** Or simply, heterogeneity. The difference in effect size among various studies.

- Heterogeneity simply tells that difference between studies are actually there and not due to chance
- So in reality, heterogeneity will always be present among studies.
- But we should test, if that is significant or not. To what extent does it affect conclusions of the meta analysis

- **Test for presence: Cochran's Q-Test**
- Cochran's Q test is the traditional test for heterogeneity in meta-analyses. Based on a chi-square (χ^2) distribution, it generates a probability that, when large, indicates larger variation across studies.

- **Quantifying heterogeneity: I^2 Test**
- The I^2 index is a more recent approach to quantify heterogeneity in meta-analyses.
- I^2 provides an estimate of the percentage of variability in results across studies that is due to real differences and not due to chance.

$I^2 = \frac{Q-df}{Q} * 100\%$	Q = Cochrane's heterogeneity stat, χ^2 distribution df = No. of studies-1

- If I^2 is 20%, this would mean that 20% of the observed variation in treatment effects cannot be attributed to chance alone
- Heterogeneity ➡ 0.25= Low | 0.5= Moderate| ≥0.75= High
- The limitation of I^2 is that it provides only a measure of global heterogeneity but **no information for the factor causing heterogeneity,** similar to Cochran's Q test.
- **Between-study variance: Tau- squared (τ^2)**

$$\tau^2 = \frac{Q-df}{Q}$$

- Tau squared is the estimate of the variance of the underlying distribution of true effect size

Measures of Heterogeneity

- **Q-statistic** (also referred to as Cochrane's Q)- It is only a measure of variation around the average and is not yet a measure of heterogeneity.
- **I^2** - Proportion of observed variance that reflects real differences in effect size-expressed as a percentage with a range from 0 to 100%.
- **τ^2** - Estimate of the variance of the true effect sizes-used to assign weights to the studies assuming Random Effect.
- **Tau (τ) -** Estimate of the standard deviation of the distribution of true effect sizes- used for computing the *prediction interval*

- **Investigating Heterogeneity: Meta regression**

- **Meta-regression** models strive to control for and explain differences in treatment effects in terms of study covariates.

- A meta-regression can be either a **linear or a logistic regression model**, and it can be based on a fixed or random effects regression.

- The unit of the analysis is the individual study included in a systematic review or meta-analysis.

- **Predictors** in the regression model are **study-level characteristics** such as study-level location, sample size, length of follow-up, drop-out rates, or study quality characteristics.

- The advantage of meta-regression is that it **determines which study-level characteristics account for heterogeneity**, rather than just providing an estimate of the global heterogeneity.

Ok, so now we have explored and estimated the heterogeneity among the studies. What if there is high heterogeneity?

- Don't pool results for meta analysis
- **Ignore** heterogeneity and use **Fixed effect model**
- **Control** for heterogeneity using **Random effect model**

- **Fixed-effect model** assumes that the effect of treatment is the same, and that variation between results in different studies is due to random error.

- Thus, a fixed-effect model can be used when the studies are considered to have the same design and methodology, or when the variability in results within a study is small, and the variance is thought to be due to random error.

- Three common methods are used for weighted estimation in a fixed-effect model:
  1) **Inverse variance-weighted estimation:** Small no. of studies with large sample size
  2) **Mantel-Haenszel estimation:** Large no. of studies with small sample size
  3) **Peto estimation**: Low event rate or one of the two groups shows zero incidence

- **Random-effect model** assumes heterogeneity between the studies being combined, and these models are used when the studies are assumed different, even if a heterogeneity test does not show a significant result.

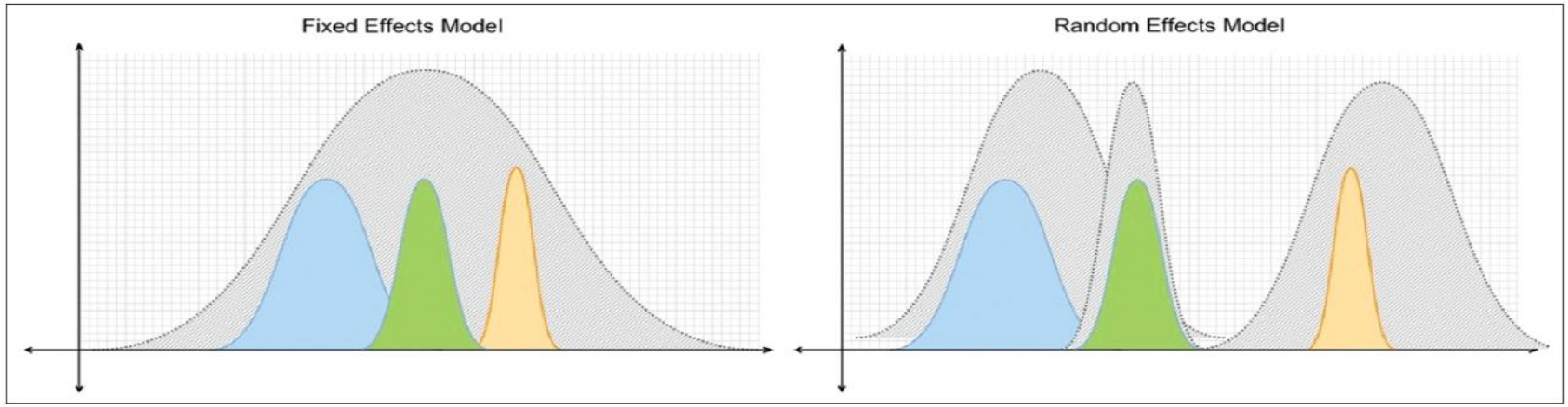
- Unlike a fixed-effect model, a random- effect model assumes that the size of the effect of treatment differs among studies.

- Thus, differences in variation among studies are thought to be due to not only random error but also between-study variability in results

- Among methods for weighted estimation in a random-effect model, the **Der Simonian and Laird method** is mostly used for dichotomous variables, while **Inverse variance-weighted estimation** is used for continuous variables

Types of Effects

Fixed Effect	Random Effect
Here we assume that there is one true effect size (hence the term fixed effect) which underlies all the studies in the analysis, and that all differences in observed effects are due to sampling error. • No Generalization	Here we allow that the true effect could vary from study to study. If it were possible to perform an infinite number of studies, the true effect sizes for these studies would be distributed about some mean. • Purpose is Generalization



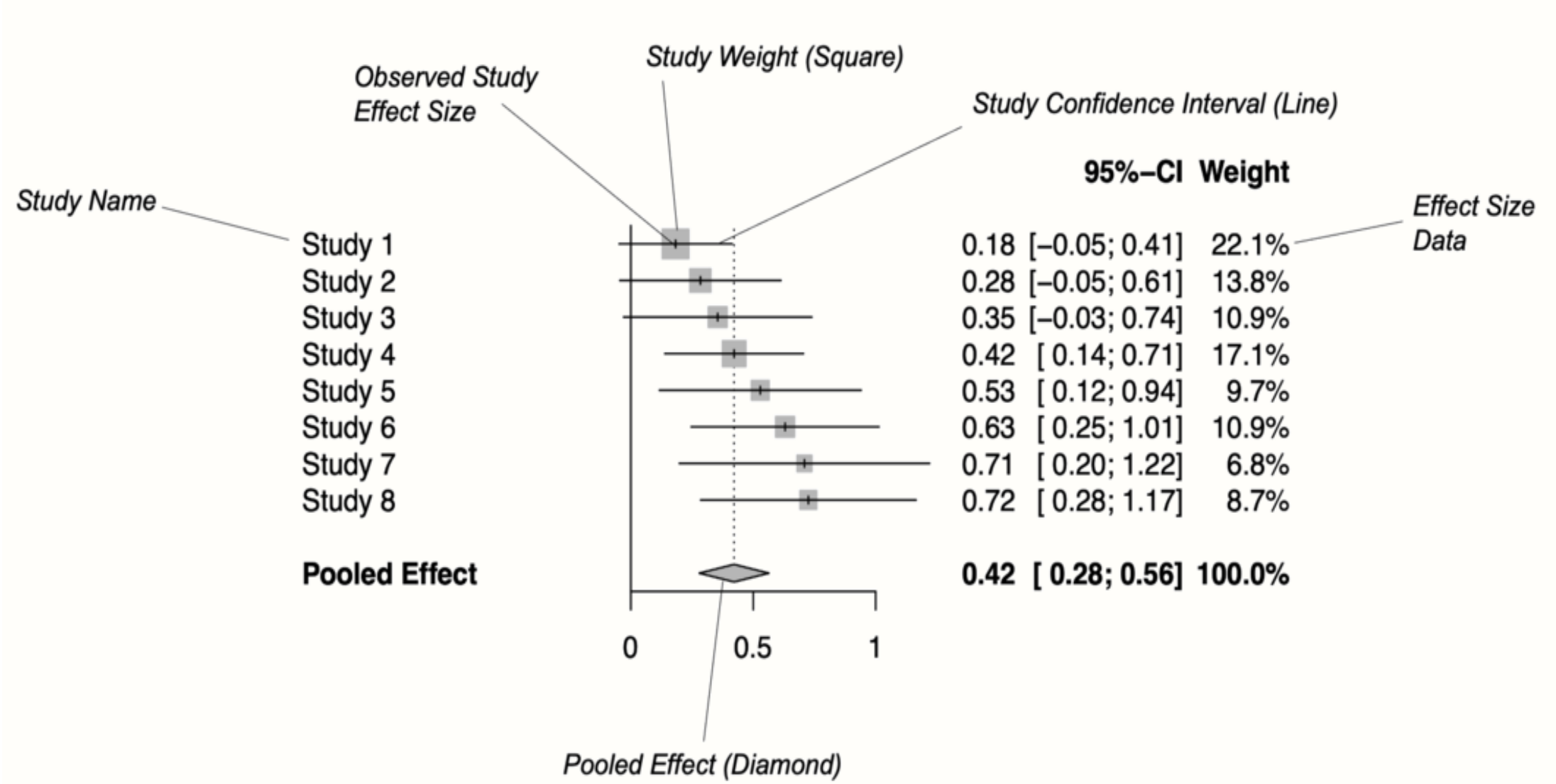
Only one true effect
Differences among study caused by chance alone, corrected with N

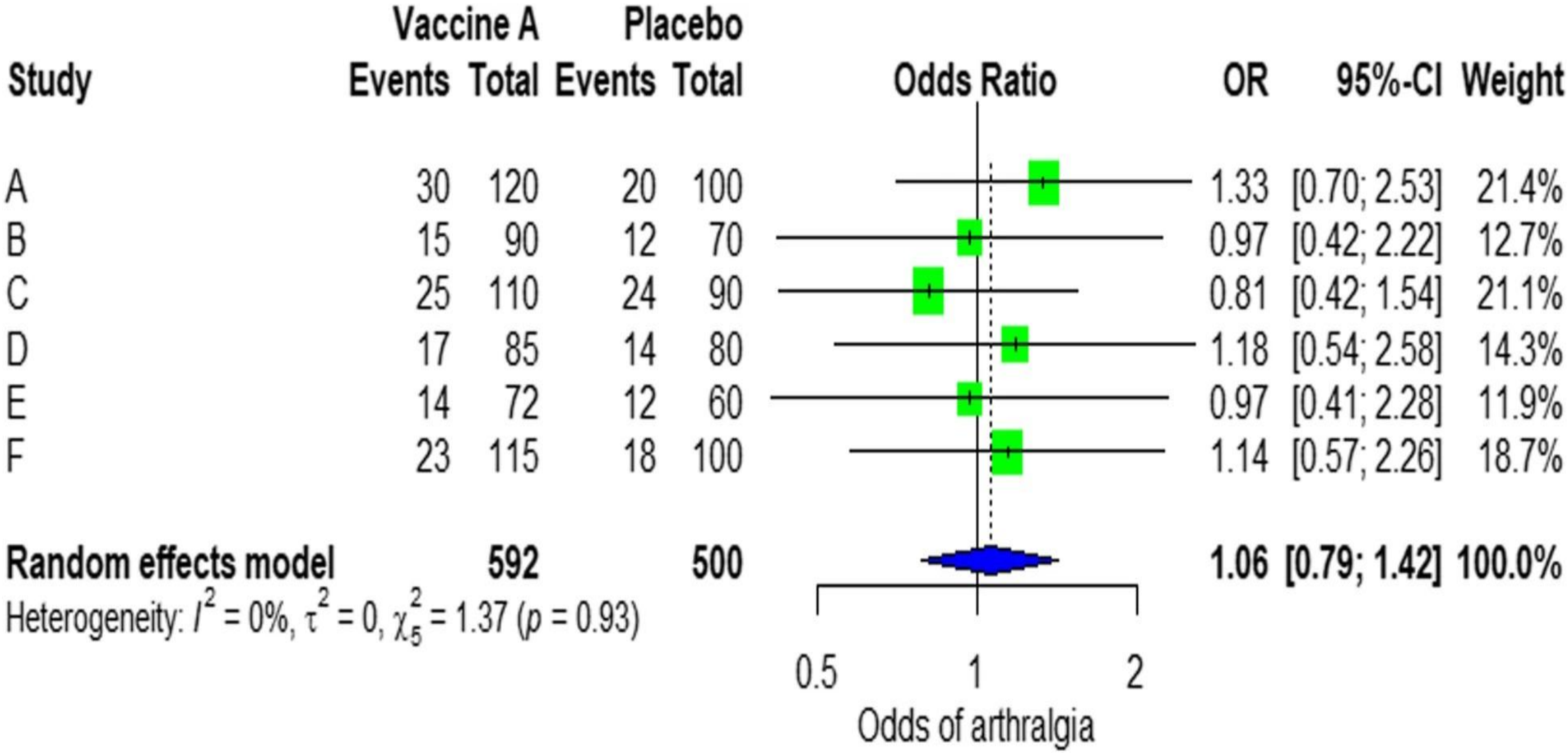
More than one true effects
Increasing the sample size could not correct

Table 2. Summary of Meta-analysis Methods Available in RevMan [28]

Type of data	Effect measure	Fixed-effect methods	Random-effect methods
Dichotomous	Odds ratio (OR)	Mantel-Haenszel (M-H) Inverse variance (IV) Peto	Der Simonian and Laird method
	Risk ratio (RR), Risk difference (RD)	Mantel-Haenszel (M-H) Inverse variance (IV)	Inverse variance (IV)
Continuous	Mean difference (MD), Standardized mean difference (SMD)	Inverse variance (IV)	Inverse variance (IV)

Forest Plot


Observed Study Effect Size
Study Weight (Square)
Study Confidence Interval (Line)
Study Name
Effect Size Data
95%–CI Weight
Study 1 0.18 [–0.05; 0.41] 22.1%
Study 2 0.28 [–0.05; 0.61] 13.8%
Study 3 0.35 [–0.03; 0.74] 10.9%
Study 4 0.42 [0.14; 0.71] 17.1%
Study 5 0.53 [0.12; 0.94] 9.7%
Study 6 0.63 [0.25; 1.01] 10.9%
Study 7 0.71 [0.20; 1.22] 6.8%
Study 8 0.72 [0.28; 1.17] 8.7%
Pooled Effect 0.42 [0.28; 0.56] 100.0%
0
0.5
1
Pooled Effect (Diamond)


Study
Vaccine A
Events Total
Placebo
Events Total
Odds Ratio
OR 95%-CI Weight
A 30 120 20 100 1.33 [0.70; 2.53] 21.4%
B 15 90 12 70 0.97 [0.42; 2.22] 12.7%
C 25 110 24 90 0.81 [0.42; 1.54] 21.1%
D 17 85 14 80 1.18 [0.54; 2.58] 14.3%
E 14 72 12 60 0.97 [0.41; 2.28] 11.9%
F 23 115 18 100 1.14 [0.57; 2.26] 18.7%
Random effects model 592 500 1.06 [0.79; 1.42] 100.0%
Heterogeneity: I^2 = 0%, τ^2 = 0, χ^2_5 = 1.37 (p = 0.93)
0.5
1
2
Odds of arthralgia

Column 1. Study details

Columns 2 and 3. The two groups being compared

Column 4. Weight

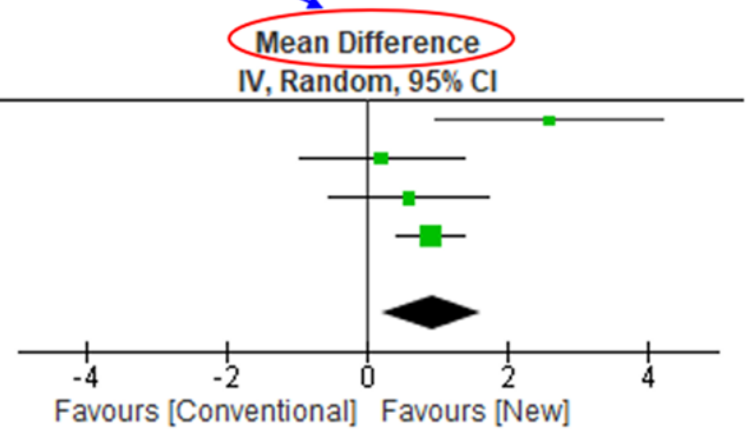
Columns 5 and 6. Study effect measure in numeric and graphic forms

A)

Study or Subgroup	Conventional Surgery Mean	SD	Total	New Surgery Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Study #1	17.1	2.1	13	14.5	2.21	14	14.0%	2.60 [0.97, 4.23]
Study #2	18.5	1.7	17	18.3	2.2	26	21.6%	0.20 [-0.97, 1.37]
Study #3	17.9	1.5	18	17.3	1.6	12	22.3%	0.60 [-0.54, 1.74]
Study #4	16.6	1.32	57	15.7	1.33	57	42.2%	0.90 [0.41, 1.39]
Total (95% CI)			**105**			**109**	**100.0%**	**0.92 [0.21, 1.63]**

Heterogeneity: Tau² = 0.25; Chi² = 5.84, df = 3 (P = 0.12); I² = 49%

Test for overall effect: Z = 2.54 (P = 0.01)

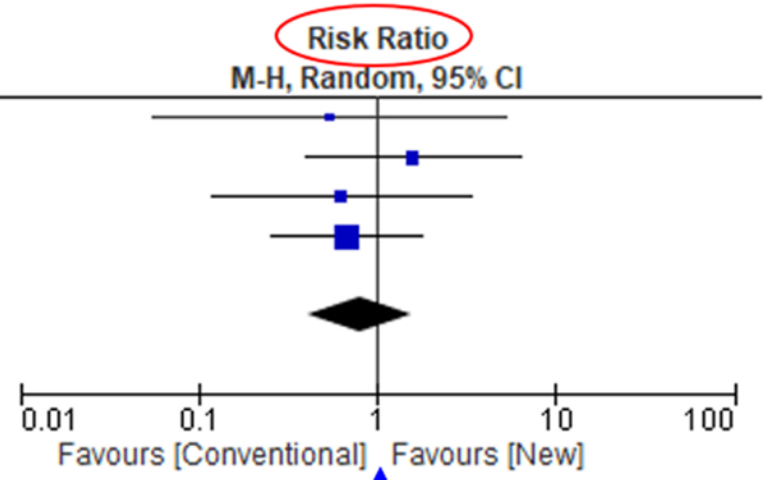


B)

Study or Subgroup	Conventional Surgery Events	Total	New Surgery Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Study #1	1	13	2	14	9.0%	0.54 [0.06, 5.26]
Study #2	4	22	3	26	24.3%	1.58 [0.39, 6.30]
Study #3	2	18	3	17	16.9%	0.63 [0.12, 3.32]
Study #4	6	57	9	57	49.9%	0.67 [0.25, 1.75]
Total (95% CI)		**110**		**114**	**100.0%**	**0.80 [0.40, 1.58]**
Total events	13		17			

Heterogeneity: Tau² = 0.00; Chi² = 1.25, df = 3 (P = 0.74); I² = 0%

Test for overall effect: Z = 0.65 (P = 0.52)



Between-study difference examined by a χ^2 test and the I^2 statistic

The vertical line: the line of no effect, with the value of 0 (continuous measure such as mean difference) or 1 (dichotomous measure such as risk ratio)

Publication Bias in Meta Analysis

- In general, a study showing a beneficial effect of a new treatment is more likely to be considered worthy of publication than one showing no effect.

- There is a considerable bias that operates at every stage of the process, with negative trials considered to contribute less to scientific knowledge than positive ones:

  - Those who conducted the study are more likely to submit the results to a peer reviewed journal;
  - Editors of journals are more likely to consider the study potentially worth publishing and send it for peer review
  - Referees are more likely to deem the study suitable for publication.

- This situation has been accentuated by two factors: first that studies have often been **too small to detect a beneficial effect even if one exists** and second that there has been **too much emphasis on 'significant' results** (i.e. $P < 0.05$ for the effect of interest).

- A proposed solution to the problem of publication bias is to **establish registers of all trials in a particular area**, from when they are funded or established.

- It is also clear that the active discouragement of studies that do not have power to detect a clinically important effect would alleviate the problem.

- Publication bias is a lesser problem for larger studies, for which there tends to be general agreement that the results are of interest, whatever they are.

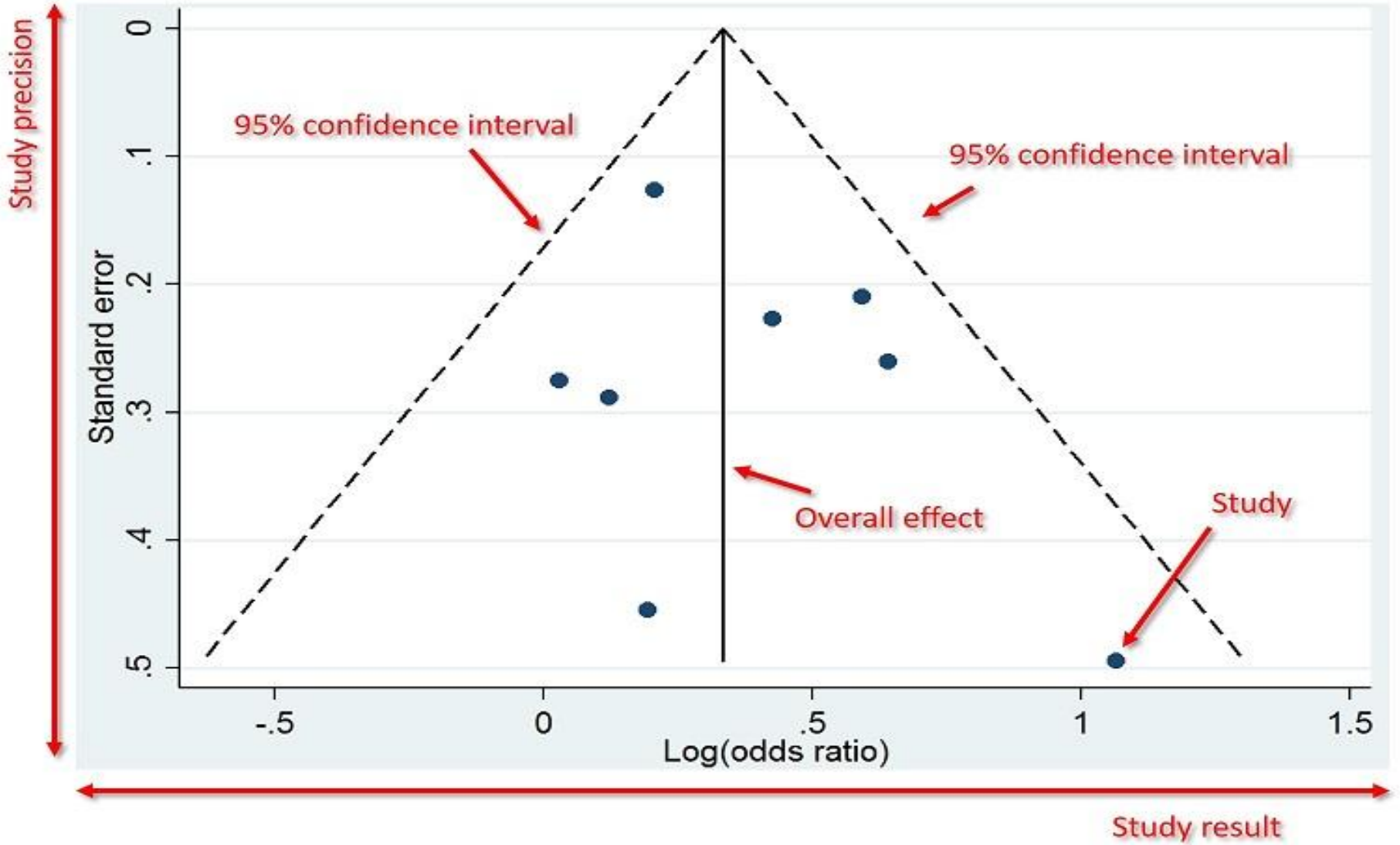
Funnel Plots to examine publication bias

- The existence of publication bias may be examined graphically by the use of ‘funnel plots’.

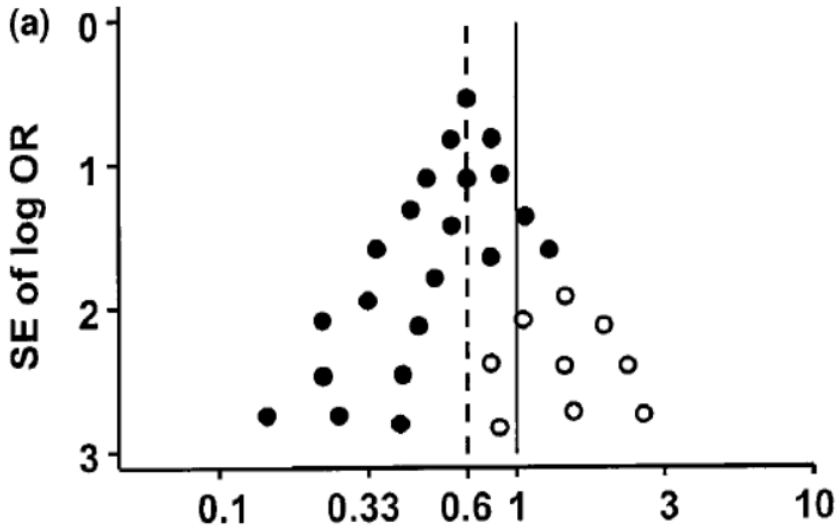
- These are simple scatter plots of the **study results/ treatment effects** on the **horizontal (x) axis** and the **precision** of that study (sample size or inverse SE) on the **vertical (y) axis**.

- The name ‘funnel plot’ is based on the fact that the **precision in the estimation of the underlying treatment effect will increase as the sample size of component studies increases.**

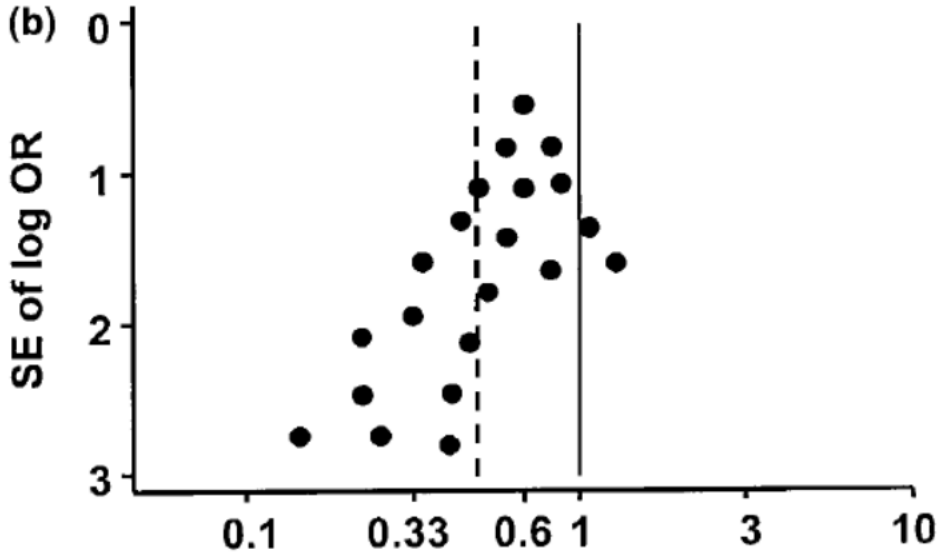
- Effect estimates from small studies will therefore scatter more widely at the bottom of the graph, with the spread narrowing among larger studies.


Study precision
Standard error
0
.1
.2
.3
.4
.5
95% confidence interval
95% confidence interval
Overall effect
Study
-.5
0
.5
1
1.5
Log(odds ratio)
Study result

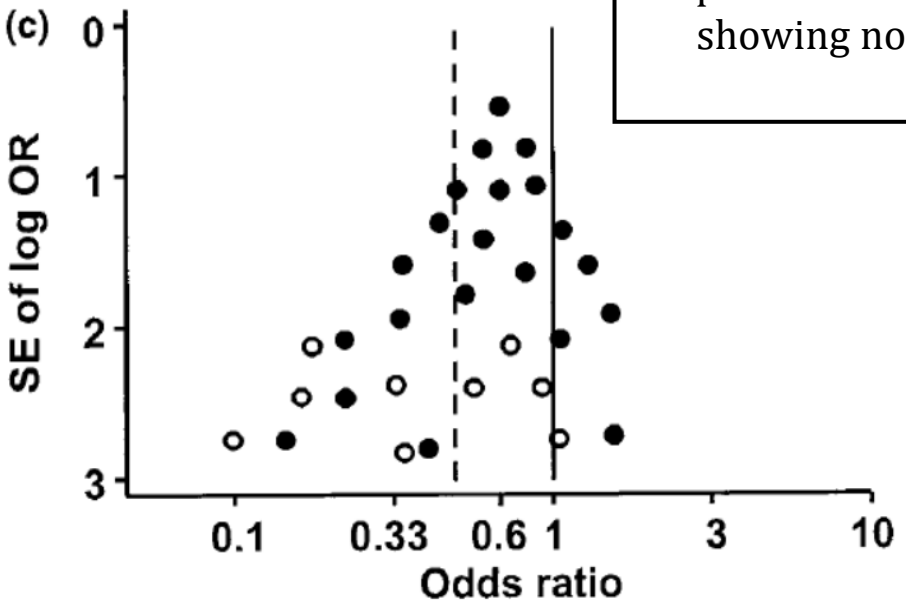
Funnel Plot



Symmetrical plot in the absence of bias (open circles indicate smaller studies showing no beneficial effects)



Asymmetrical plot in the presence of publication bias (smaller studies showing no beneficial effects are missing)



Asymmetrical plot in the presence of bias due to low methodological quality of smaller studies (open circles indicate small studies of inadequate quality whose results are biased towards larger beneficial effects

- **Relative measures of treatment effect** (risk ratios or odds ratios) are plotted on a logarithmic scale.

- This is important to ensure that effects of the same magnitude but opposite directions, for example risk ratios of 0.5 and 2, are equidistant from 1 (corresponding to no effect).

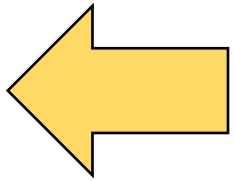
- However, the **statistical power of a trial** is determined both by the total sample size and the number of participants developing the event of interest.

- For example, a study with 100,000 patients and 10 events is less likely to show a statistically significant effect of a treatment than a study with 1000 patients and 100 events.

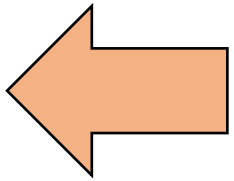
- The standard error of the effect estimate, rather than total sample size, has therefore been increasingly used in funnel plots

Softwares for Meta-Analysis:

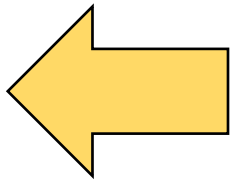
- RevMan (Cochrane)
- Metafor (R package)
- Comprehensive Meta-Analysis Software (CMA)
- Jamovi
- MetaXL (Excel add-on)
- MetEasy (Excel add-on)



Free



Freemium



Free

Conclusion

- Systematic reviews and meta-analysis (the quantitative analysis of such reviews) are now accepted as an important part of medical research.

- While the analytical methods are relatively simple, there is still controversy over appropriate methods of analysis.

- Systematic reviews are substantial undertakings, and those conducting such reviews need to be aware of the potential biases which may affect their conclusions.

- However, the explosion in medical research information and the availability of reviews on-line mean that synthesis of research findings in form of Systematic reviews and Meta-analysis is likely to be of ever increasing importance to the practice of medicine.

Registering your protocol for Systematic Review and Meta-analysis

Why register?

- Identify if any reviews already in progress or published
- Registration provides transparency in the review process.
- Improves quality and increase confidence
- It helps counter publication bias
- Safeguard against reporting biases



Photo by Marc Schulte on Unsplash

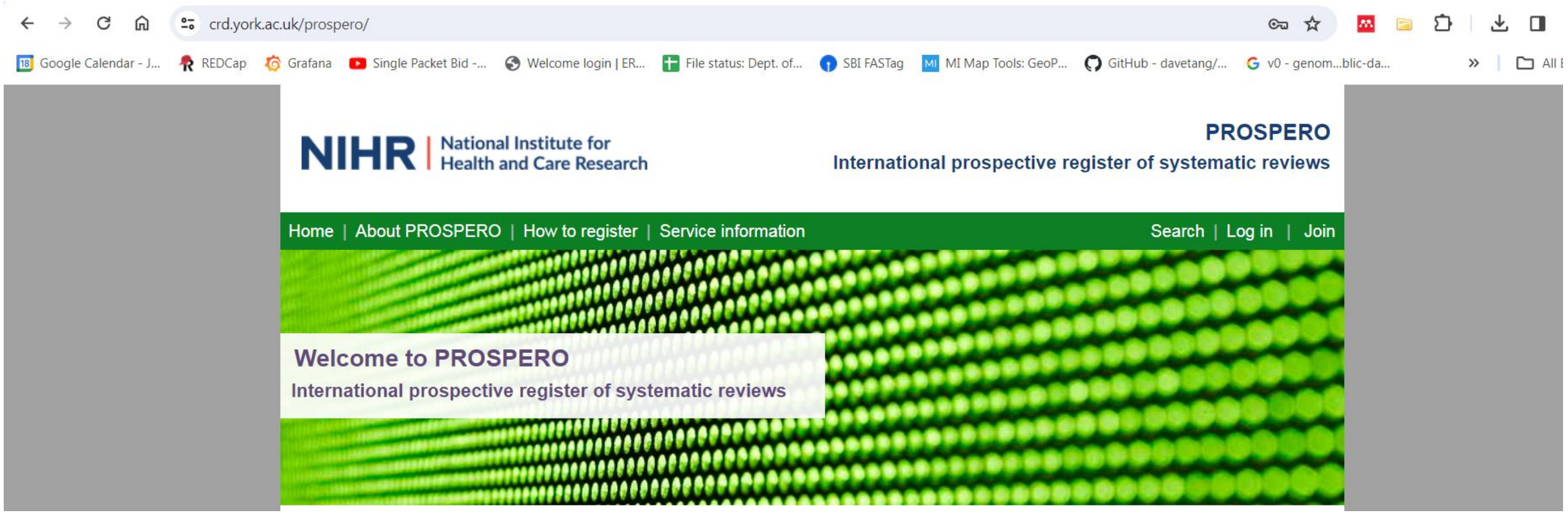
What is the purpose of prospective registration?

Increasing transparency and reducing the potential for bias in systematic reviews.

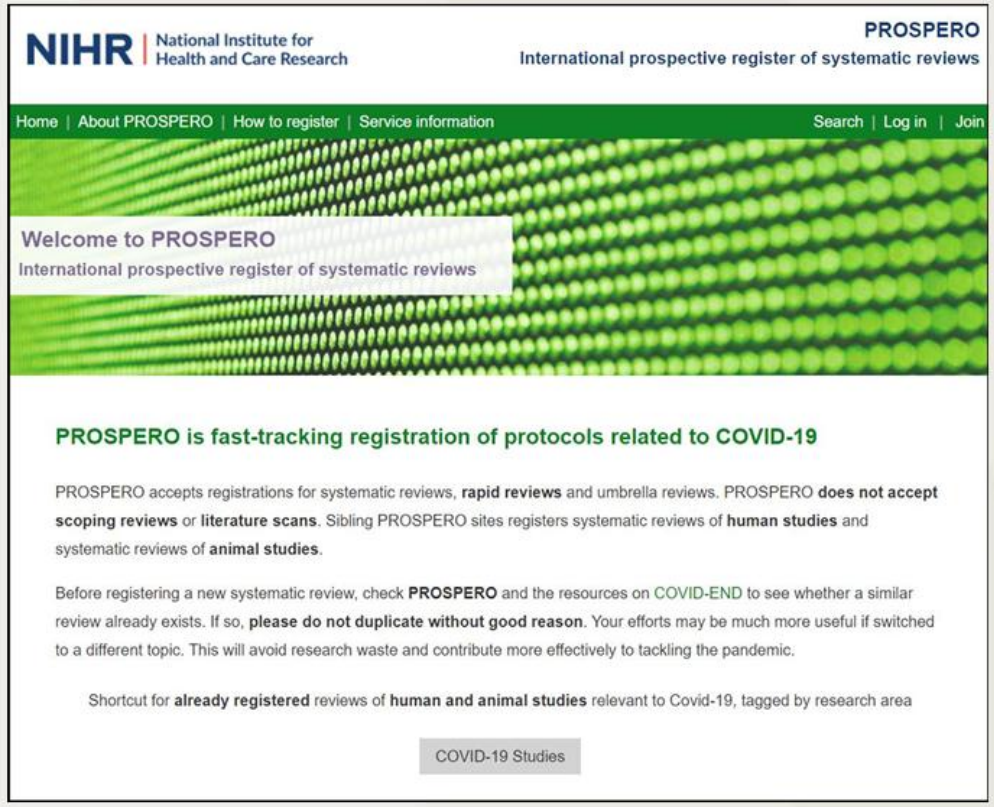
PROSPERO
International prospective register of systematic reviews

Where do we find PROSPERO?

https://www.crd.york.ac.uk/prospero/

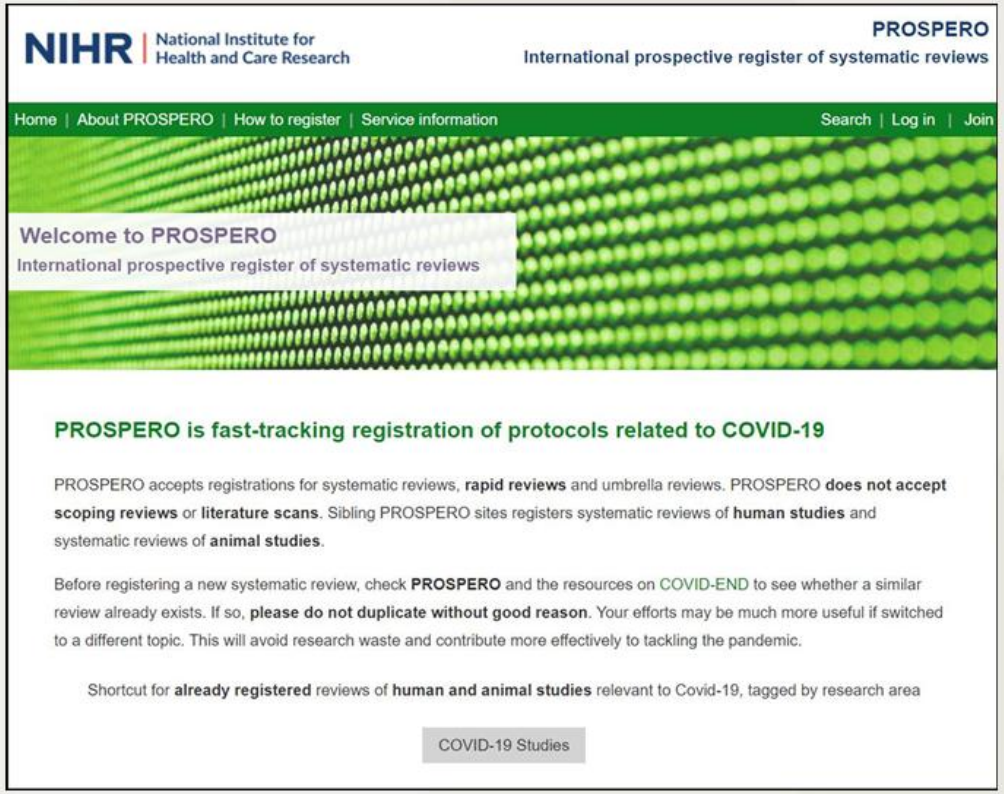


International Prospective Register of Systematic Reviews (PROSPERO)



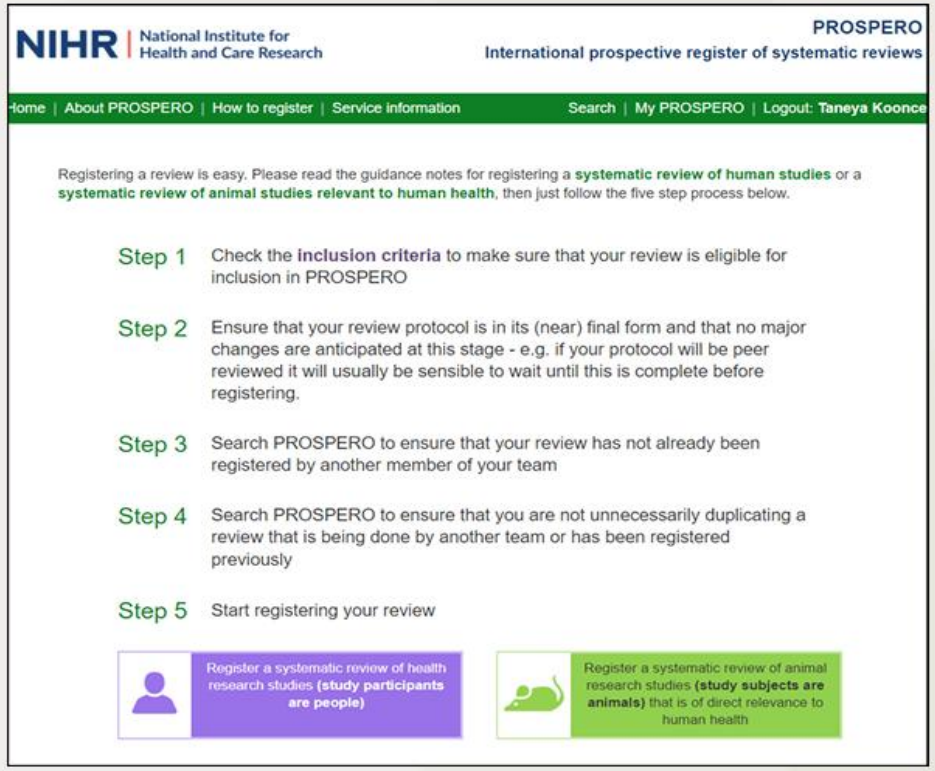
- Created in 2011
- Funded by the National Institute of Health and Care Research
- No cost to register
- Accepts rapid and umbrella reviews with a health-related outcome
- Submissions undergo quality review; processing can take 3-6 months
- The *largest* and *longest-running* database for systematic review protocol registration (more than 100,000 protocols)

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PROSPERO provides researchers with detailed instructions for protocol structure.



NIHR | National Institute for Health and Care Research

PROSPERO
International prospective register of systematic reviews

Home | About PROSPERO | How to register | Service information | Search | My PROSPERO | Logout: **Taneya Koonce**

Registering a review is easy. Please read the guidance notes for registering a **systematic review of human studies** or a **systematic review of animal studies relevant to human health**, then just follow the five step process below.

Step 1 Check the **inclusion criteria** to make sure that your review is eligible for inclusion in PROSPERO

Step 2 Ensure that your review protocol is in its (near) final form and that no major changes are anticipated at this stage - e.g. if your protocol will be peer reviewed it will usually be sensible to wait until this is complete before registering.

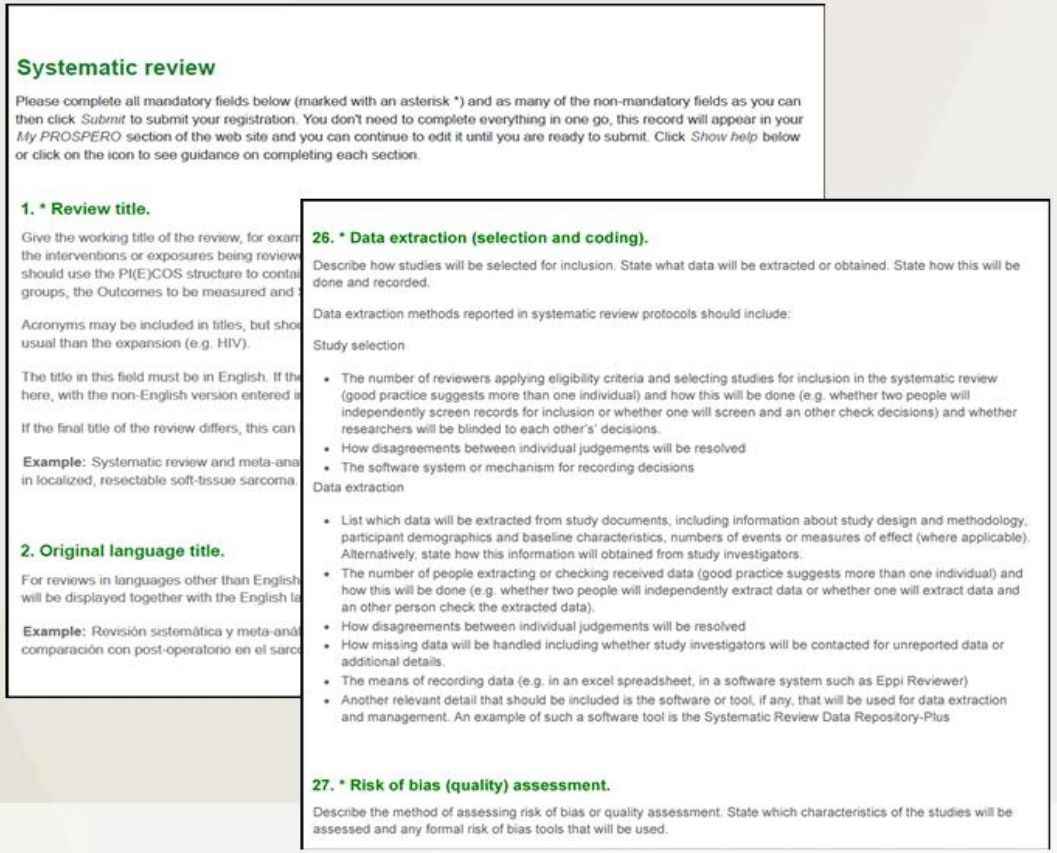
Step 3 Search PROSPERO to ensure that your review has not already been registered by another member of your team

Step 4 Search PROSPERO to ensure that you are not unnecessarily duplicating a review that is being done by another team or has been registered previously

Step 5 Start registering your review

Register a systematic review of health research studies **(study participants are people)**

Register a systematic review of animal research studies **(study subjects are animals)** that is of direct relevance to human health



Systematic review

Please complete all mandatory fields below (marked with an asterisk *) and as many of the non-mandatory fields as you can then click *Submit* to submit your registration. You don't need to complete everything in one go, this record will appear in your *My PROSPERO* section of the web site and you can continue to edit it until you are ready to submit. Click *Show help* below or click on the icon to see guidance on completing each section.

1. * Review title.

Give the working title of the review, for exam
the interventions or exposures being review
should use the PI(E)COS structure to conta
groups, the Outcomes to be measured and

Acronyms may be included in titles, but sho
usual than the expansion (e.g. HIV).

The title in this field must be in English. If th
here, with the non-English version entered i

If the final title of the review differs, this can

Example: Systematic review and meta-ana
in localized, resectable soft-tissue sarcoma.

2. Original language title.

For reviews in languages other than English
will be displayed together with the English la

Example: Revisión sistemática y meta-aná
comparación con post-operatorio en el sarc

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

Data extraction methods reported in systematic review protocols should include:

Study selection

- The number of reviewers applying eligibility criteria and selecting studies for inclusion in the systematic review (good practice suggests more than one individual) and how this will be done (e.g. whether two people will independently screen records for inclusion or whether one will screen and an other check decisions) and whether researchers will be blinded to each other's' decisions.
- How disagreements between individual judgements will be resolved
- The software system or mechanism for recording decisions

Data extraction

- List which data will be extracted from study documents, including information about study design and methodology, participant demographics and baseline characteristics, numbers of events or measures of effect (where applicable). Alternatively, state how this information will obtained from study investigators.
- The number of people extracting or checking received data (good practice suggests more than one individual) and how this will be done (e.g. whether two people will independently extract data or whether one will extract data and an other person check the extracted data).
- How disagreements between individual judgements will be resolved
- How missing data will be handled including whether study investigators will be contacted for unreported data or additional details.
- The means of recording data (e.g. in an excel spreadsheet, in a software system such as Eppi Reviewer)
- Another relevant detail that should be included is the software or tool, if any, that will be used for data extraction and management. An example of such a software tool is the Systematic Review Data Repository-Plus

27. * Risk of bias (quality) assessment.

Describe the method of assessing risk of bias or quality assessment. State which characteristics of the studies will be assessed and any formal risk of bias tools that will be used.

Before registration at PROSPERO?
Go through PRISMA guidelines

PRISMA
TRANSPARENT REPORTING OF SYSTEMATIC REVIEWS AND META-ANALYSES

HOME | PRISMA STATEMENT | EXTENSIONS | TRANSLATIONS | PROTOCOLS | ENDORSEMENT | News

Welcome to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) website!

PRISMA is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses. PRISMA primarily focuses on the reporting of reviews evaluating the effects of interventions, but can also be used as a basis for reporting systematic reviews with objectives other than evaluating interventions (e.g. evaluating aetiology, prevalence, diagnosis or prognosis).

Who should use PRISMA?

- Authors: PRISMA aims to help authors improve the reporting of systematic reviews and meta-analyses.
- Journal Peer reviewers and editors: PRISMA may also be useful for critical appraisal of published systematic reviews, although it is not a quality assessment instrument to gauge the quality of a systematic review.

News Feed

Key Documents

- PRISMA 2020 Checklist
- PRISMA 2020 flow diagram
- PRISMA 2020 Statement
- PRISMA 2020 Explanation and Elaboration

PROSPERO
International prospective register of systematic reviews

equator network

Before registration at PROSPERO?

Read a related Systematic Review

PLOS Digital Health

RESEARCH ARTICLE

A systematic review of the limitations of large language models in generating healthcare content

Mohsen Khosravi[1*], Zahra Zamaninasab[2], Seyyed Morteza Mojtabaeian[3], Emine Kübra Dindar Demiray[4], Morteza Arab-Zozani[1]

1 Social Determinants of Health Research Center, Birjand University of Medical Sciences, Birjand, Iran, **2** Department of Epidemiology and Biostatistics, School of Health, Social Determinants of Health Research Center, Birjand University of Medical Sciences, Birjand, Iran, **3** Department of Healthcare Services Management, School of Management and Medical Informatics, Shiraz University of Medical Sciences, Shiraz, Iran, **4** Department of Infection Diseases and Clinical Microbiology, Siirt University Medical School, Siirt, Türkiye

* mohsenkhosravi@live.com

Before registration at PROSPERO?

Search PubMed, PROSPERO, etc to make sure that a similar review has not been published recently

There are relatively few systematic review on the topic

Before registration at PROSPERO?
Make sure data extraction has not started

NIHR | National Institute for Health and Care Research

PROSPERO
International prospective register of systematic reviews

Home | About PROSPERO | How to register | Service information | Search | Log in | Join

Help with registration

Accessing and completing the registration form

What happens after submitting a form

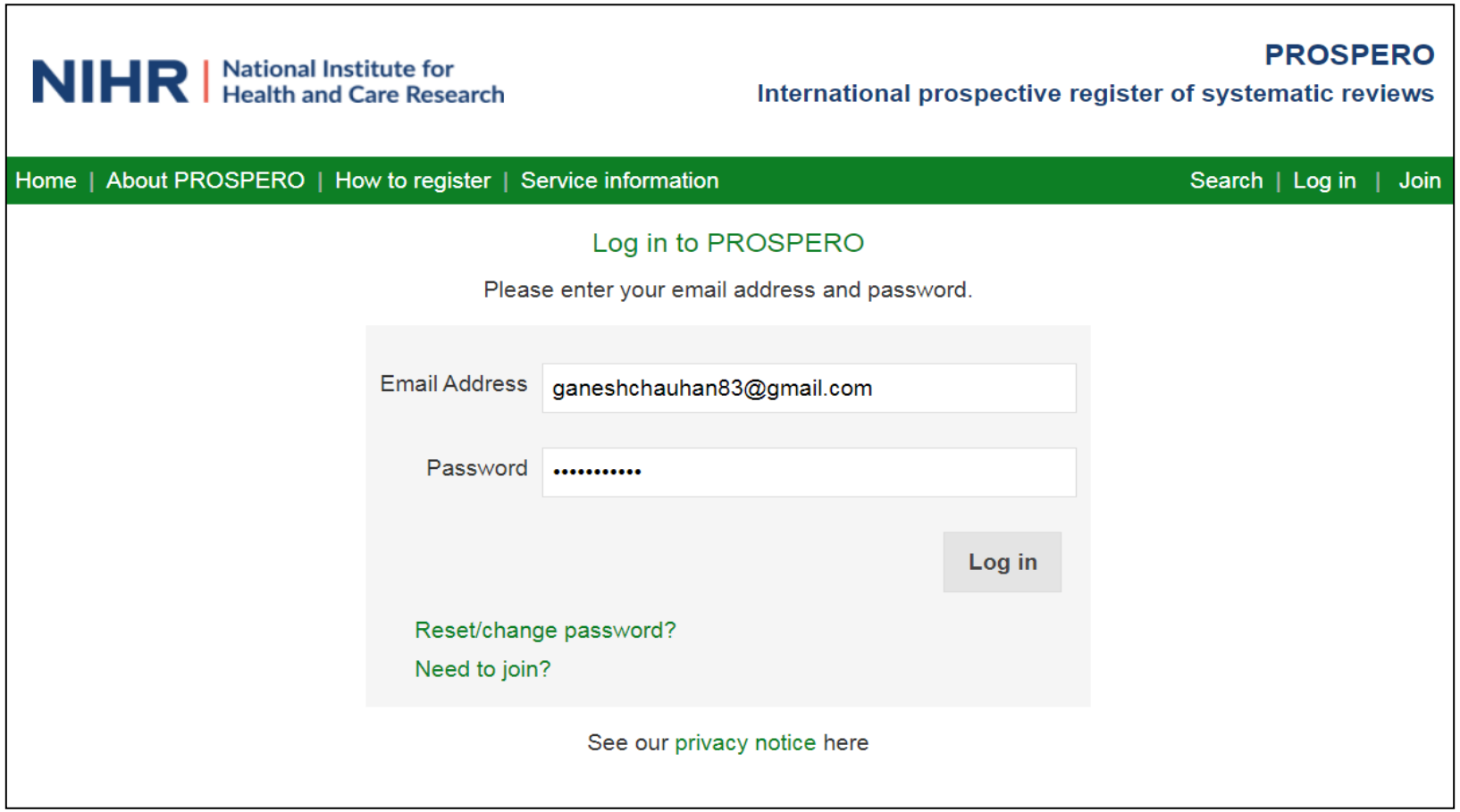
Making changes, amendments and updating a published

From October 2019 PROSPERO will require earlier registration.

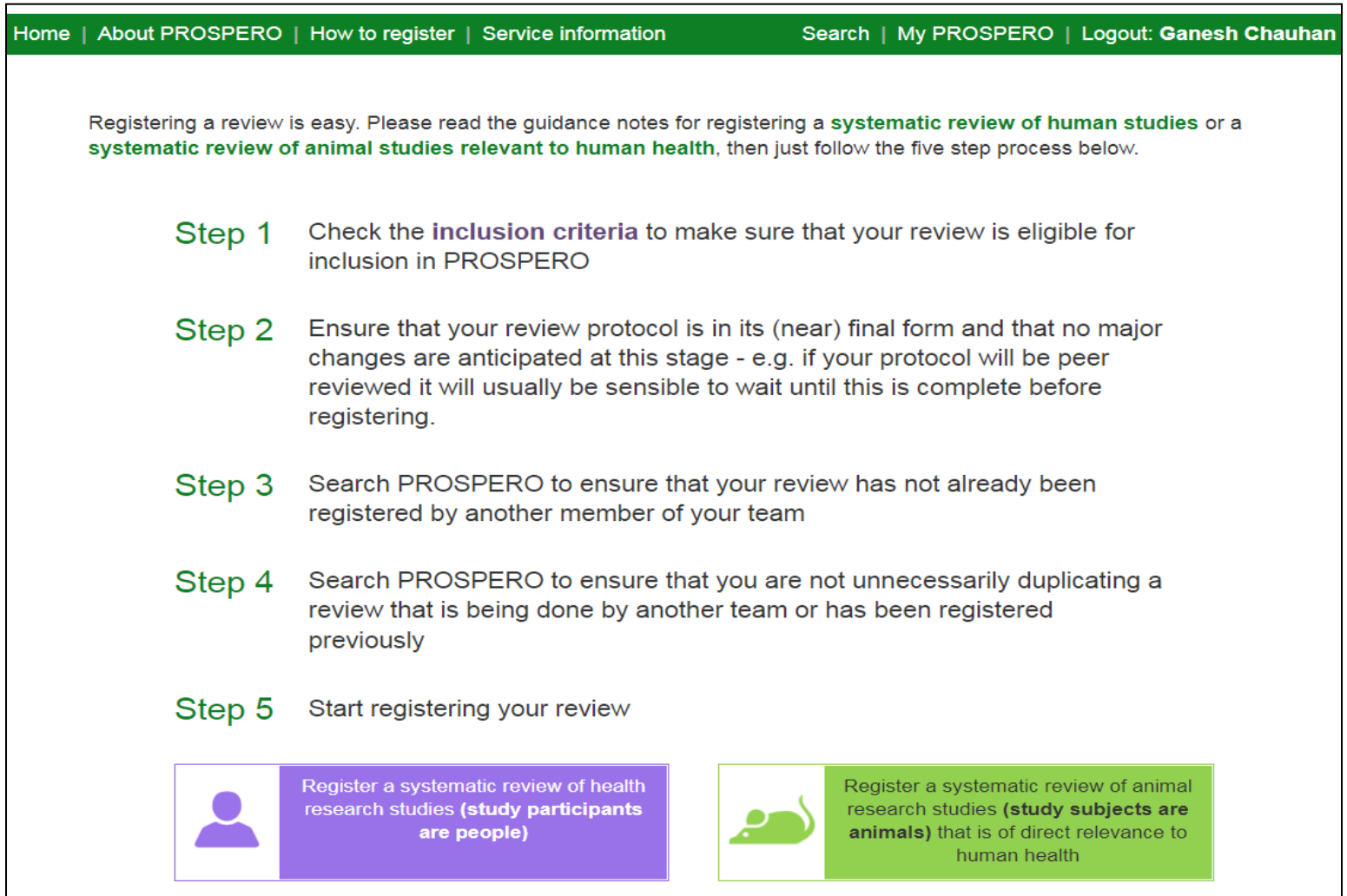
Previously, we accepted registrations provided the reviewers had not *completed* their data extraction but, **from 1st October 2019, we will only accept reviews provided that data extraction has not yet *started*.**

We hope that this will reduce the potential for bias by reducing the opportunity for data extraction to lead to conscious or subconscious effects on the review, for instance by influencing decisions to include or exclude certain items in order to shape a review so that it reaches a desired conclusion.

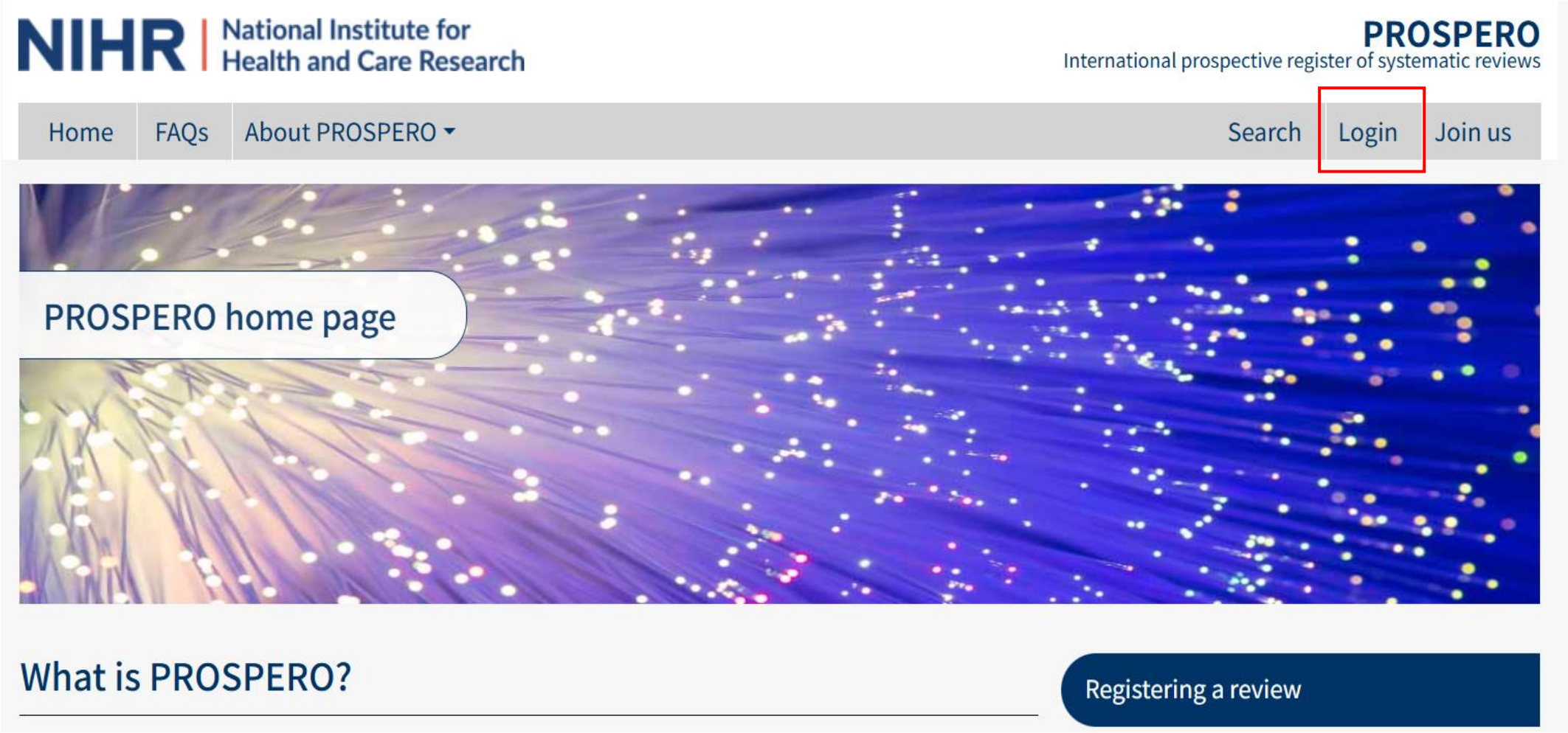
Before registration at PROSPERO?
Create a login on Prospero website



Are you ready for registration?



Steps for registration



NIHR | National Institute for Health and Care Research

Home | FAQs | About PROSPERO

Welcome to PROSPERO

Please enter your details

Login with your ORCID

or

Email
raoww14@gmail.com

Password
• • • • • • • • • • • •

Forgotten password?

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Don't have an account? Join us today

PROSPERO
International prospective register of systematic reviews

Search | My PROSPERO | Logout: Wenwang Rao



Registering a review

Registering a review is quick and easy and free of charge. You must be logged in to register a review.

Register your review now

Registering your review

Terms of use for registering reviews

1. Registration in PROSPERO is governed by the Terms and Conditions of Use and the Privacy policy agreed when creating your PROSPERO account.
2. All data entered and information provided must be accurate and truthful.
3. Record owners are responsible for all content within their record(s).
4. Reviews must be in English and in scope (see Eligibility for inclusion for more details).
5. Reviews **must be registered before data extraction has started**, or before individual participant data (IPD) have been received.
6. Reviews building on previous work should state this and should not plagiarise published work by reproducing text.
7. PROSPERO accepts information in good faith and relies on the integrity of those registering their reviews, who are entirely responsible for content.
8. Action will be taken if inaccuracies in data are identified at any time including information supplied about the stage of review or anticipated completion time. Deliberate falsification of data including review team membership will be regarded as research misconduct.
9. The PROSPERO team carry out random checks and reserve the right to withdraw registrations if they are found to be out of scope or contain information known to be inaccurate
10. **You MUST NOT create a registration for a completed review even if a journal asks you for a registration number**
11. The PROSPERO team does not have resources to create, update or amend records on behalf of record owners.

Click here for more information...

☐ I have read and understand the terms above and agree to abide by them

Continue

What are you registering?

PROSPERO uses different templates for different types of review.

Use this page to select the correct template. This page will determine the information you need to provide in your registration record. Answer correctly here to avoid having to restart the process later.

Review of human or animal studies?

PROSPERO comprises two complementary registries - one of reviews of human health and wellbeing and one of animal pre-clinical studies that have a clear and direct link to human health and wellbeing

I am registering

- ○ A systematic review of studies with **people/humans as participants**
- ○ A systematic review of studies with **animals as participants** that is directly relevant to human health

Is this a live registration record or a training record?

(Help me decide)

- This record is NOT being created as part of a student project or for training purposes. It WILL be published on the public site.

  All of the registration process will be available to you and your record will be published on the public PROSPERO site once all team members approve publication.

- This record is being created for training purposes. It **WILL NOT** be published on the public site.

  *All of the registration process will be available to you but **your record will not be published on the public PROSPERO site**. You will be able to give access to your record to your supervisor. It is not possible to change this status at a later date - if you decide you want the record to be published you will have to create a new record.*

What type of review are you planning?

- Intervention review

Assessing the benefits, harms or experience of a health intervention (medicine, treatment, vaccine, device, preventative measure, procedure or policy).

- Coming soon: Diagnostic test accuracy

Assessing the accuracy of a diagnostic test in detecting or excluding a particular disease or condition.

- Coming soon: Prognostic review

Assessing the probable course or future outcome of people with a particular health problem.

- Coming soon: Review of incidence or prevalence

Assessing the prevalence or incidence of a disease or condition.

- Coming soon: Methodology

Examining the evidence on methodological aspects of systematic reviews, trials and other evaluations of health and social care.

Will this be a living systematic review?

A living systematic review (LSR) is a continuously updated, dynamic summary of research that stays current by regularly incorporating new evidence as it emerges after publication. These reviews use ongoing literature surveillance and frequent updates.

If you are not sure whether this review will be a Living Review, choose "standard systematic review" now - you can change that later.

- (●) **This review will be a standard systematic review**
- ( ) This review will be a living systematic review

Summary

You are creating a registration record for an **intervention review**, a systematic review of studies with **people/humans as participants**. This will be a **standard** systematic review.

This record is NOT being created as part of a student project or for training purposes. It WILL be published on the public site.

Register your review now

This record is being created for training purposes. It **WILL NOT** be published on the public site.

*All of the registration process will be available to you but **your record will not be published on the public PROSPERO site.** You will be able to give access to your record to your supervisor. It is not possible to change this status at a later date - if you decide you want the record to be published you will have to create a new record.*

Summary

You are creating a registration record for an **intervention review**, a systematic review of studies with **people/humans as participants**. This will be a **standard** systematic review.

This record is being created for training purposes. It WILL NOT be published on the public site.

It is not possible to change this status at a later date - if you decide you want the record to be published you will have to create a new record.

Register your review now

A network meta-analysis of non-pharmacological interventions for improving sleep quality in adolescents with bipolar disorder.

Complete all sections below. You will not be able to submit your record until all sections are complete

- REVIEW TITLE AND BASIC DETAILS
- ELIGIBILITY CRITERIA
- SIMILAR REVIEWS
- TIMELINE OF THE REVIEW
- AVAILABILITY OF FULL PROTOCOL
- SEARCHING AND SCREENING
- DATA COLLECTION PROCESS
- OUTCOMES TO BE ANALYSED
- PLANNED DATA SYNTHESIS
- CURRENT REVIEW STAGE
- REVIEW AFFILIATION, FUNDING AND PEER REVIEW
- ADDITIONAL INFORMATION

Preview this PROSPERO record | Go to record dashboard

Print Close preview

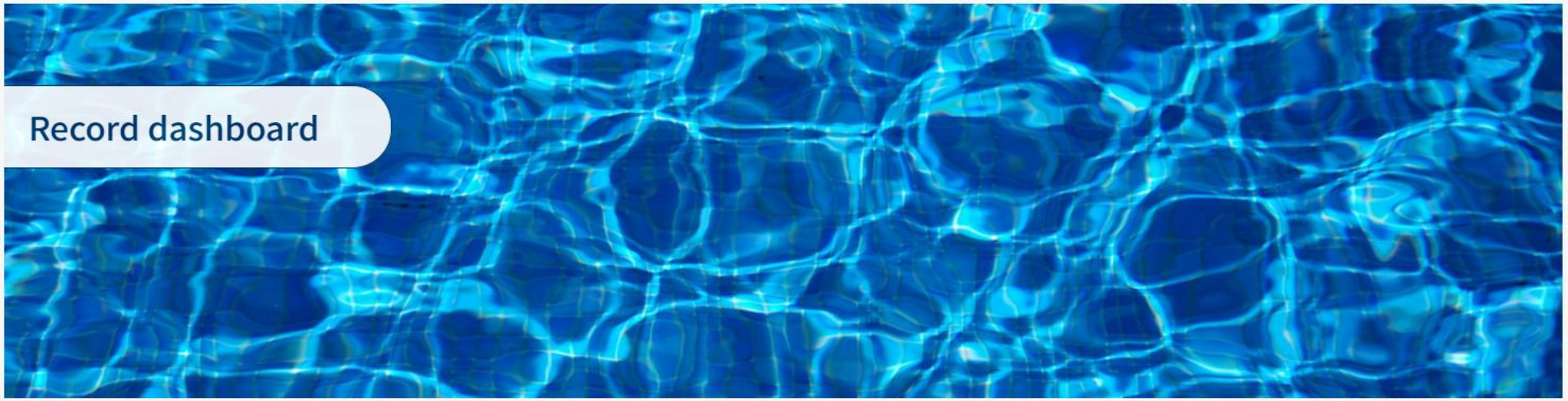
A network meta-analysis of non-pharmacological interventions for improving sleep quality in adolescents with bipolar disorder.

No team members

REVIEW TITLE AND BASIC DETAILS

Review title

A network meta-analysis of non-pharmacological interventions for improving sleep quality in adolescents with bipolar disorder.



A network meta-analysis of non-pharmacological interventions for improving sleep quality in adolescents with bipolar disorder.

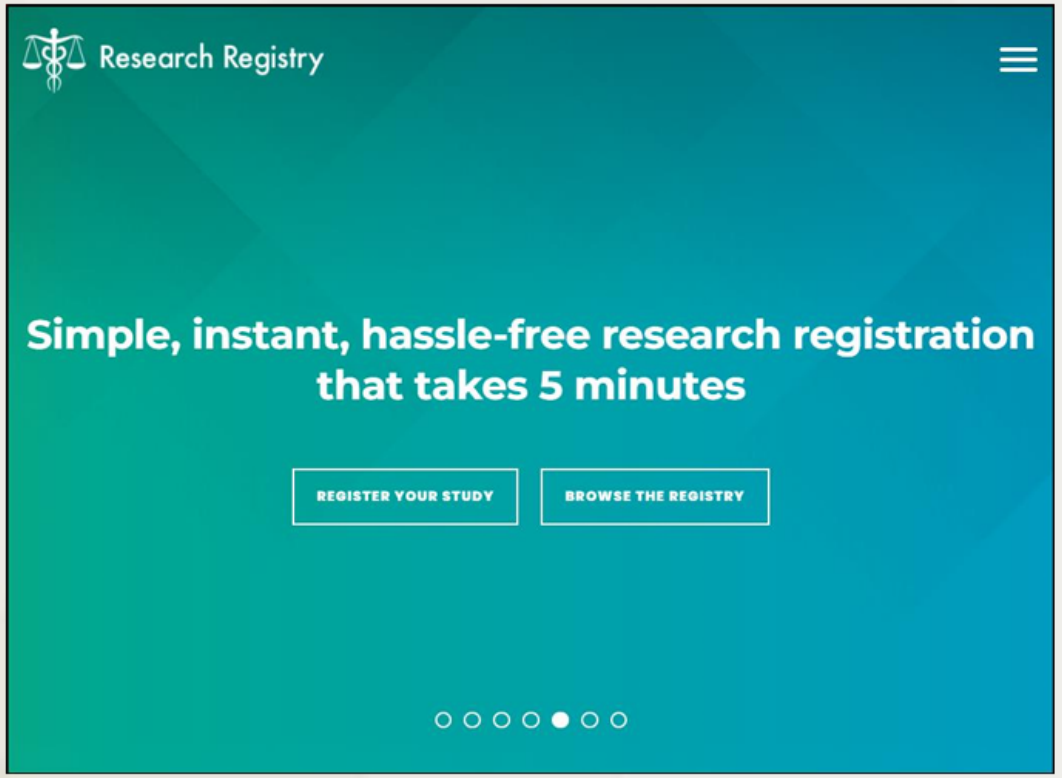
Editing status	Registration status	This version
In process 28% done	Not yet registered	Version 1.0 (Template: People v2)

How to Register Your Protocol With PROSPERO

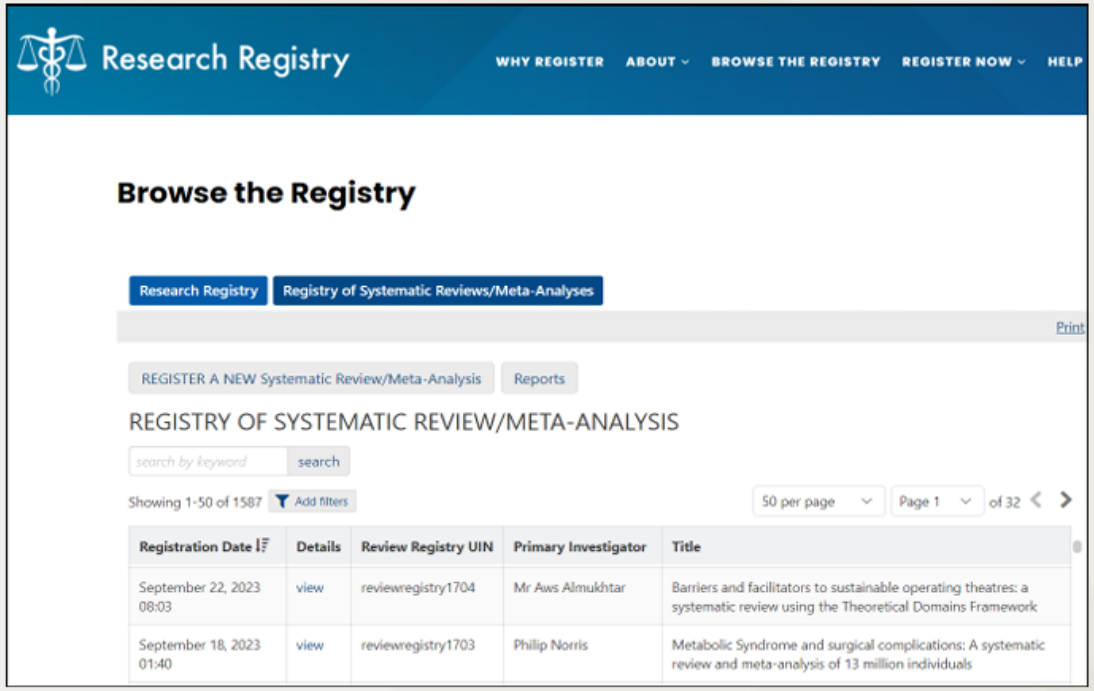


Research Registry – Registry of Systematic Reviews/Meta Analyses

- Started in 2015
- Operated by the International Journal of Surgery Publishing Group and the IDEAL consortium
- Current registration cost: 99£
- Accepts any kind of research protocol; special section for systematic reviews and meta-analyses
- Data curators perform basic submission criteria checks (e.g., check for duplicates, eliminate animal studies); protocols are published immediately
- Contains around 9,000 protocols; of which approximately 1,600 are systematic reviews (as of October 2023)

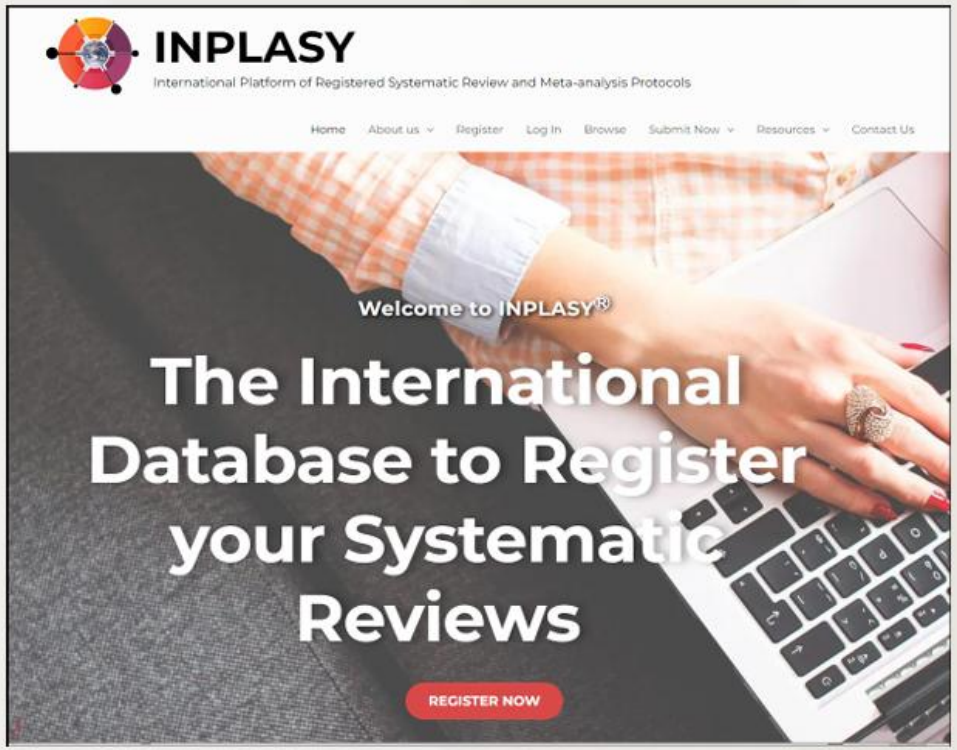


Registration is structured, though no field-by-field guide is provided.

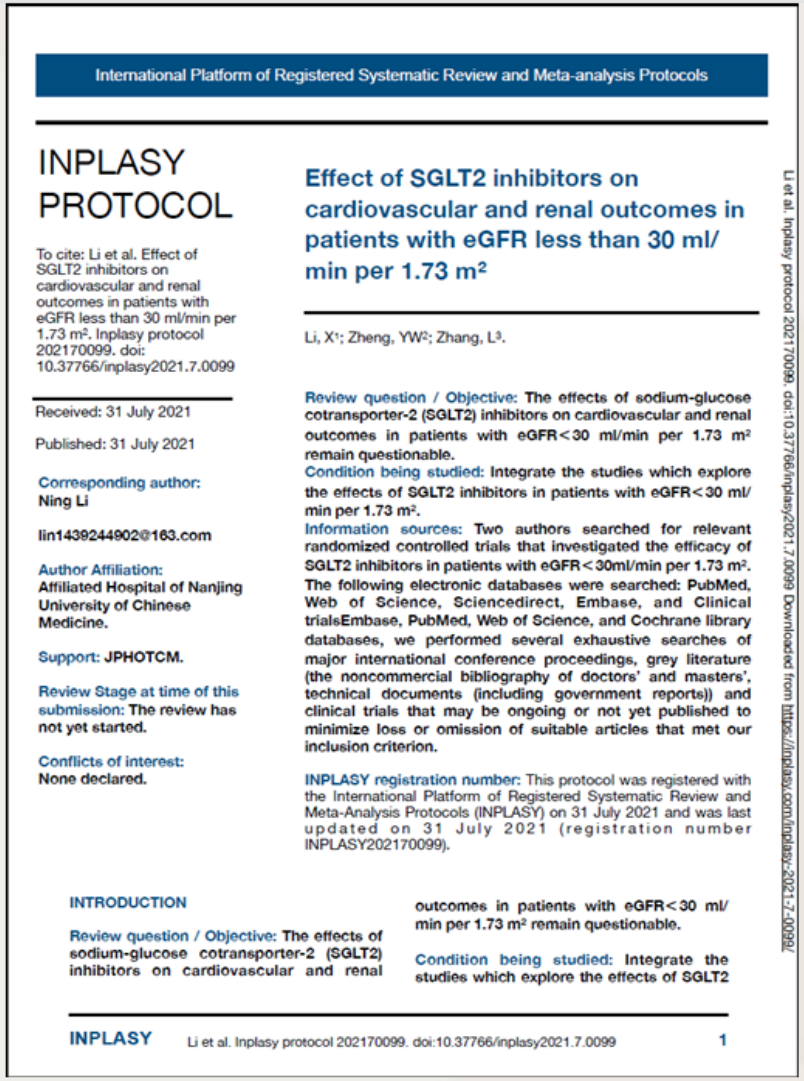




International Platform of Registered Systematic Review and Meta-Analysis Protocols (INPLASY)



- Launched in 2020; site operated by author submission fees
- Registration cost: $20 with fees for protocol updates ($9)
- Accepts systematic, rapid, scoping, and mapping reviews
- Submissions undergo basic quality review; protocols are published within 48 hours.
- Unique digital object identifiers (DOIs) are assigned to each protocol
- Crossref integration with ORCID for updating author research profiles
- Site contains ~5,500 systematic review protocols (as of October 2023)



International Platform of Registered Systematic Review and Meta-analysis Protocols

INPLASY PROTOCOL

To cite: Li et al. Effect of SGLT2 inhibitors on cardiovascular and renal outcomes in patients with eGFR less than 30 ml/min per 1.73 m². Inplasy protocol 202170099. doi: 10.37766/inplasy2021.7.0099

Received: 31 July 2021

Published: 31 July 2021

Corresponding author:
Ning Li

lin1439244902@163.com

Author Affiliation:
Affiliated Hospital of Nanjing University of Chinese Medicine.

Support: JPHOTCM.

Review Stage at time of this submission: The review has not yet started.

Conflicts of interest:
None declared.

Effect of SGLT2 inhibitors on cardiovascular and renal outcomes in patients with eGFR less than 30 ml/min per 1.73 m²

Li, X[1]; Zheng, YW[2]; Zhang, L[3].

Review question / Objective: The effects of sodium-glucose cotransporter-2 (SGLT2) inhibitors on cardiovascular and renal outcomes in patients with eGFR<30 ml/min per 1.73 m² remain questionable.
Condition being studied: Integrate the studies which explore the effects of SGLT2 inhibitors in patients with eGFR<30 ml/min per 1.73 m².
Information sources: Two authors searched for relevant randomized controlled trials that investigated the efficacy of SGLT2 inhibitors in patients with eGFR<30ml/min per 1.73 m². The following electronic databases were searched: PubMed, Web of Science, Sciencedirect, Embase, and Clinical trialsEmbase, PubMed, Web of Science, and Cochrane library databases, we performed several exhaustive searches of major international conference proceedings, grey literature (the noncommercial bibliography of doctors' and masters', technical documents (including government reports)) and clinical trials that may be ongoing or not yet published to minimize loss or omission of suitable articles that met our inclusion criterion.

INPLASY registration number: This protocol was registered with the International Platform of Registered Systematic Review and Meta-Analysis Protocols (INPLASY) on 31 July 2021 and was last updated on 31 July 2021 (registration number INPLASY202170099).

INTRODUCTION

Review question / Objective: The effects of sodium-glucose cotransporter-2 (SGLT2) inhibitors on cardiovascular and renal outcomes in patients with eGFR<30 ml/min per 1.73 m² remain questionable.

Condition being studied: Integrate the studies which explore the effects of SGLT2

Li et al. Inplasy protocol 202170099. doi:10.37766/inplasy2021.7.0099 Downloaded from https://inplasy.com/inplasy-2021-7-0099/

Protocols are formatted as PDF documents, and each follows PRISMA-P recommendations.

INPLASY protocol includes recommended items

PRISMA-P recommendations

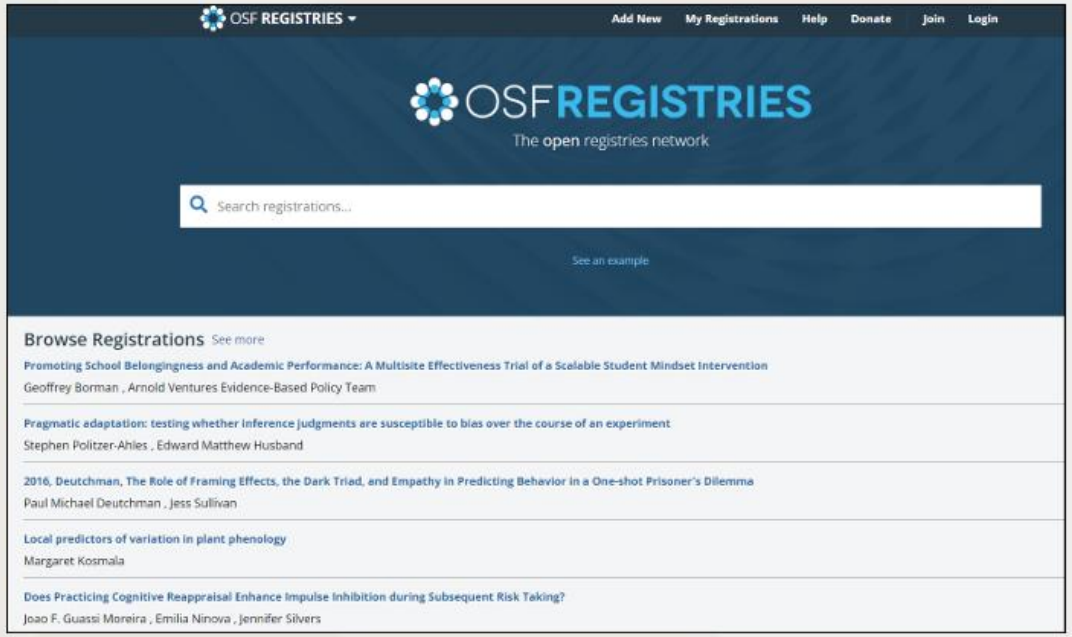
25	15	17
Section/topics	Items	Potential benefits

The International Platform of Registered Systematic Review and Meta-analysis Protocols (INPLASY®) at two years: an analysis of 3,082 registered protocols on inplasy.com, platform features, and website statistics (2023 preprint)

- Submissions from 45 countries
- 80% of registrations were systematic reviews or meta-analyses
- 21.2% of registrations subsequently published in scientific journals

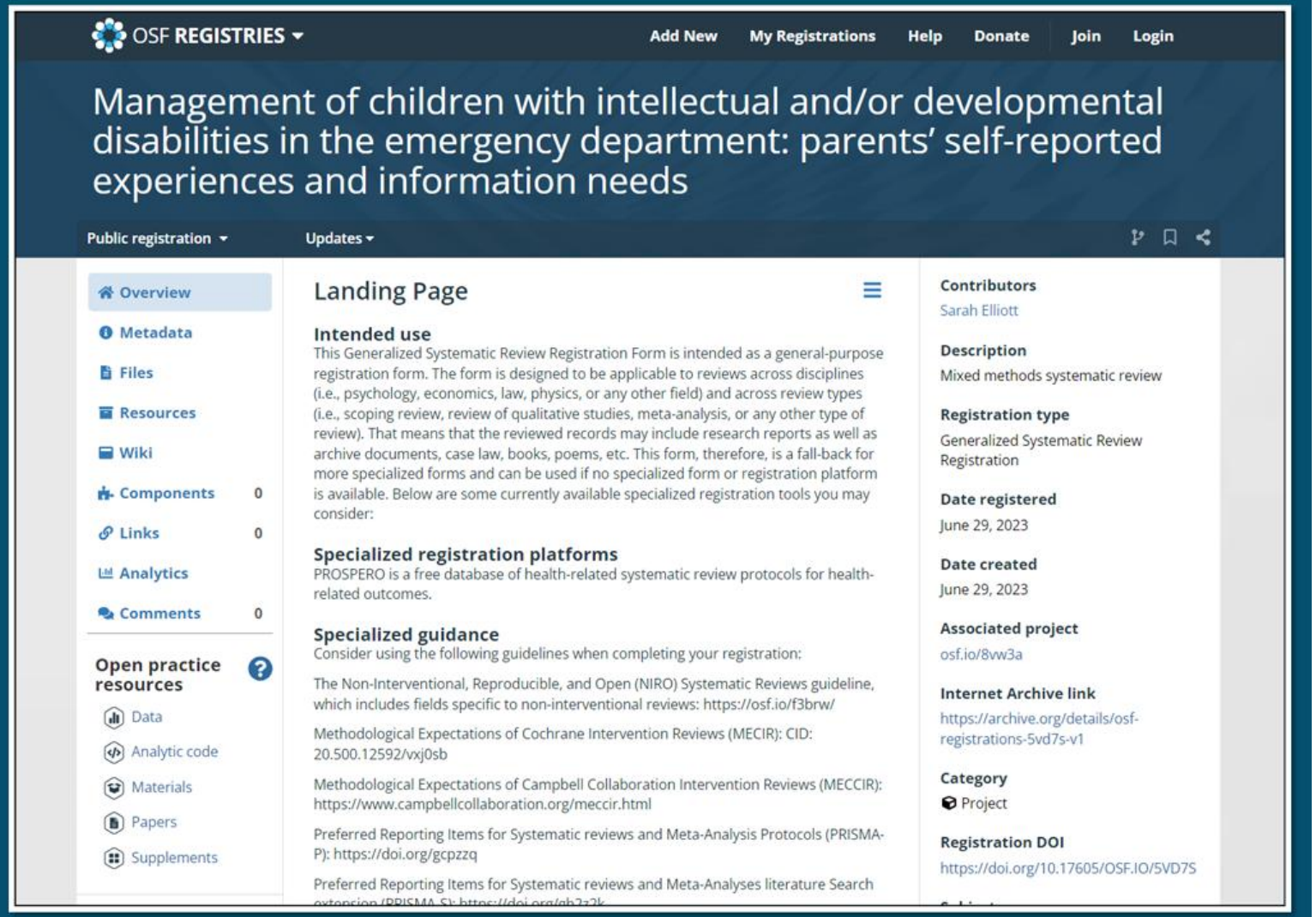
Open Science Framework Registry

- Started in 2013
- OSFRegistries is part of open science network (i.e,. OSF, OSF Preprints, OSF Meetings, OSF Institutions)
- No cost to register protocols
- Accepts any kind of protocol submission
- Submissions published within a few days; no quality review provided
- Approximately 15,000 systematic and scoping reviews (as of October 2023)



***Registered protocols are*:**

- Well-structured
- Contain multiple metadata fields
- Links can be made with data files, associated study materials, published articles



van den Akker OR, Peters GY, Bakker CJ, Carlsson R, Coles NA, Corker KS, Feldman G, Moreau D, Nordström T, Pickering JS, Riegelman A, Topor MK, van Veggel N, Yeung SK, Call M, Mellor DT, Pfeiffer N. Increasing the transparency of systematic reviews: presenting a generalized registration form. Syst Rev. 2023 Sep 22;12(1):170. doi: 10.1186/s13643-023-02281-7. PMID: 37736736; PMCID: PMC10514995.

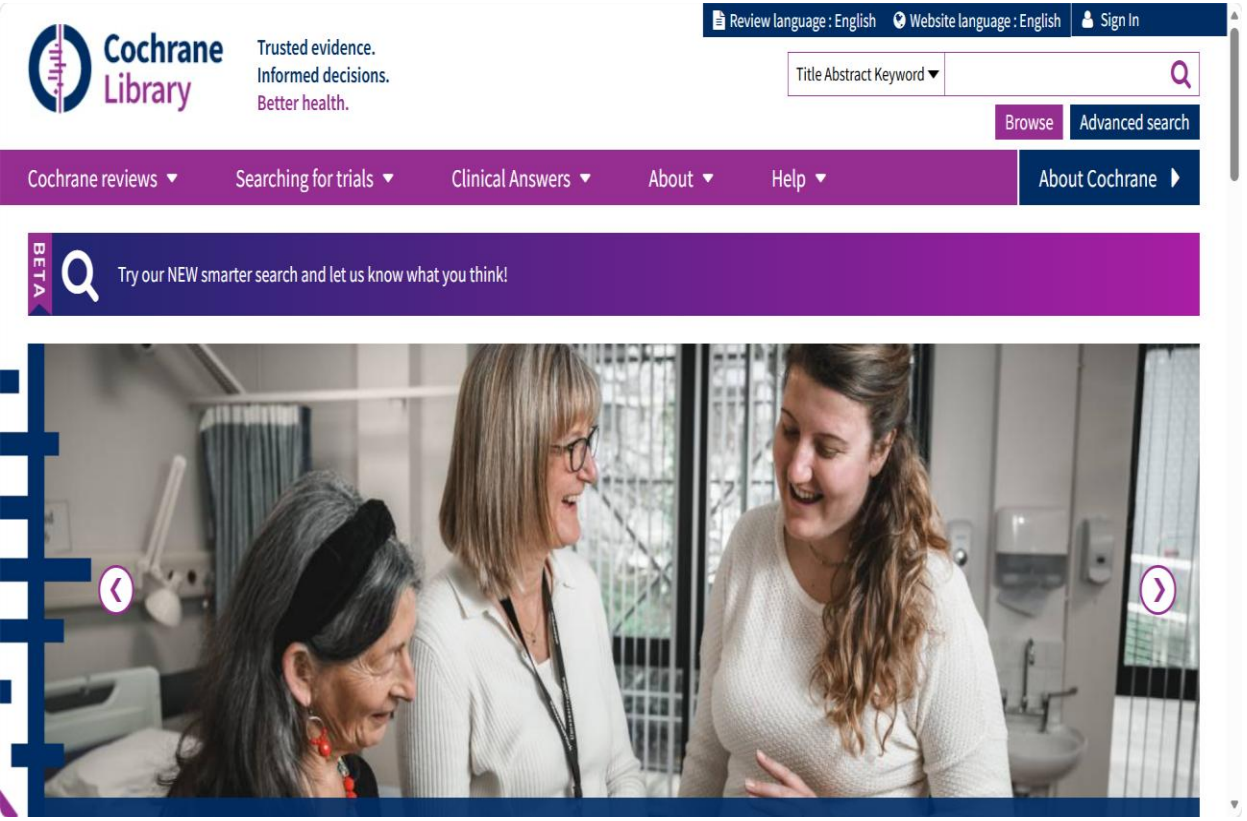
Selecting a protocol registration platform depends on your study team's goals and needs.

- **PROSPERO -** is largest and most established registry; not all types of reviews accepted; time delays are significant
- **Open Science Framework -** is second largest repository; provides structured registration form, quick publication turnaround time
- **INPLASY & Research Registry** – both have fees; smaller repositories

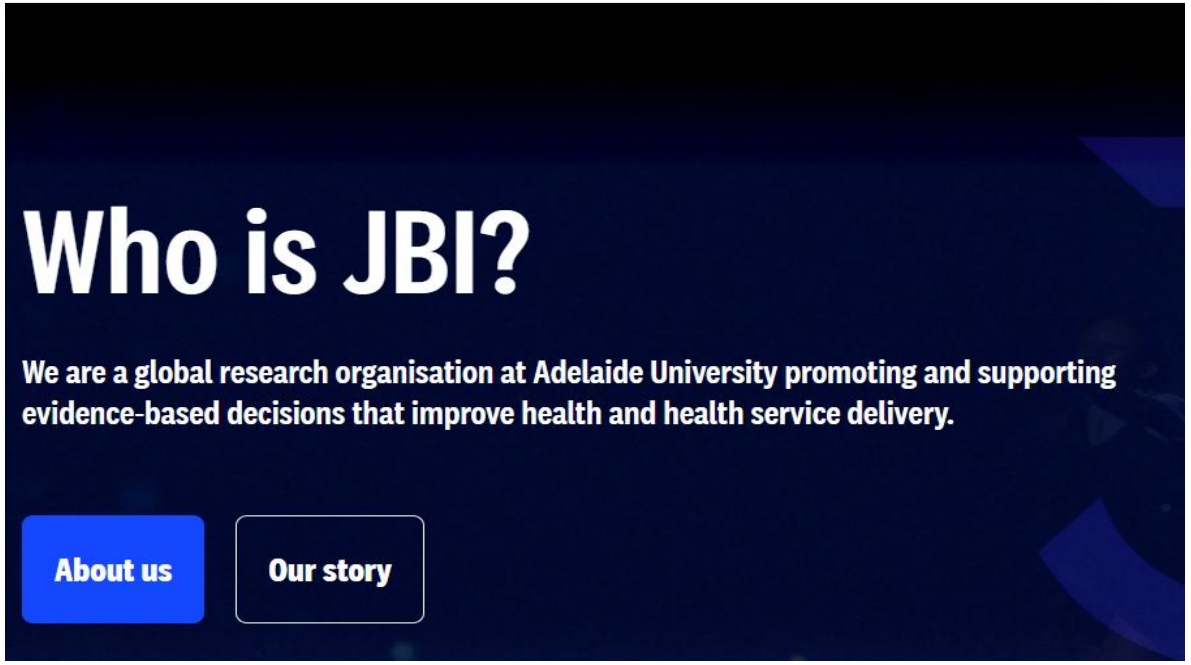


Image by Arek Socha from Pixabay

Professional registration platform



Very rigorous review process



JBI affiliated entities only

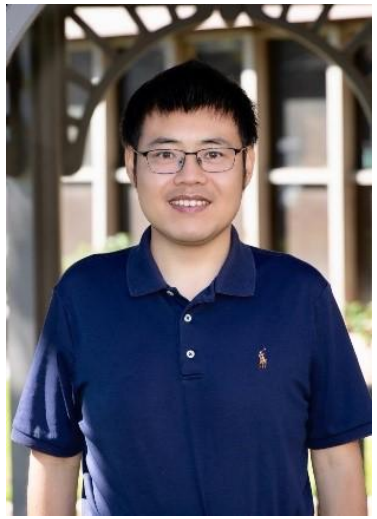


The first assignment will be to complete the registration of a protocol !!!

Assignment will be released soon.


Thank you!

Team for AI-Empowered Smart Healthy Ageing





Visiting Student Recruitment

For Juniors/Seniors & 1st-Year Graduates







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Associate Professor
Tel: (853) 8599 3824
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Qiu Huang
Associate Professor
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Hsuan Yu Lu, James
Lecturer
Tel:(853) 8599 3825
Email: hylu@mpu.edu.mo

Wenwang Rao
Lecturer
Tel:(853) 8599 3826
Email: raoww@mpu.edu.mo

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